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A systematic literature review of fintech, financial inclusion and vulnerable groups

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Abstract

This paper follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses protocol to transparently and systematically search, screen and review 185 peer-reviewed journal articles on fintech and the financial inclusion of vulnerable groups. These articles were published in 136 indexed journals before 2025, involving 160 authors and affiliations from 64 countries. Descriptive statistics show that research focusing on women, poverty and rural communities has received much attention, whereas research concerning the elderly and people with disabilities remains limited. Additionally, there has been a significant upward trend in both the number of publications and citations in this field in recent years. The literature generally suggests that while fintech has the potential to facilitate financial inclusion for vulnerable groups, persistent challenges remain, particularly in addressing geographic, demographic and socio-economic inequalities. The limitations and recommendations of previous studies, and four key research gaps along with corresponding future research directions are outlined. First, the growing threats and risks associated with the use of fintech, such as digital financial fraud, cybersecurity breaches, and system vulnerabilities. Second, the need for more research on fintech and financial inclusion among the elderly, people with disabilities and migrants, especially the challenges and impacts of digital financial inclusion. Third, the interaction and bi-directional relationships, and the integration or substitution of fintech and traditional finance within the context of vulnerable groups. Finally, the role and function of regulatory frameworks, and regulatory and supervisory mechanisms in promoting digital financial inclusion for vulnerable groups amid the rapid growth and evolution in fintech.

Keywords: Elderly; Entrepreneur; Financial Literacy; Migrant, People with disabilities; Prisma; Psalsar; Rural; SDG; Women

1. Introduction

The digital transformation within the financial services sector, facilitated by financial technology or better known as fintech, has improved business operational efficiency, customers' experience, and financial inclusion. It broadens access to

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innovative financial solutions, ranging from digital payments and currencies to crowdfunding and peer-to-peer (P2P) lending, thereby fostering greater equity and efficiency. By leveraging technologies like mobile platforms and blockchain, fintech has reformed the financial ecosystem and make financial services available and affordable not only to traditional clients but also to the vulnerable groups, including those who were previously excluded due to geographic, socioeconomic, and financial barriers such as the elderly, women, poor, ethnic minorities, people with disabilities, rural, and marginalized populations [1-2]. Research has also shown that fintech drives financial inclusion and accelerates the realisation of United Nations' sustainable development goals (SDGs) [3-5], particularly the no poverty goal (SDG 1), gender equality goal (SDG 5), and reduced inequalities goal (SDG 10), which aim to eradicate poverty in all its forms everywhere, promote women's empowerment to achieve gender equality, and reduce inequality within and among countries [6]. Improved accessibility to financial services enables and facilitates individuals, particularly the vulnerable groups to engage in secure savings, obtain loans, make investments, and pursue entrepreneurial activities. This empowerment, in turn, strengthens resilience against economic uncertainties, protects them from poverty, and subsequently improves their quality of life and well-being [7-9].

While fintech generally promotes financial inclusion and reduces gender inequities, income inequalities, and socio-economic disparities, the presence of heterogeneity within and across communities and countries is still evident. Although the World Bank's Global Findex Database 2025 report shows that there was an 5% increase in account ownership worldwide during the period of 2021-2024 owing to fintech such as mobile money accounts; there are approximately 1.3 billion adults worldwide (21%) still experience financial exclusion, especially among women (23%), and the poor (28%), who neither own a financial institution account nor a mobile money account [10]. While global financial inclusion has progressed in recent years, with overall account ownership worldwide increasing from 74% to 79% from 2021 to 2024, these benefits have not been distributed equally across all populations. As shown in Figure 1, the gaps in account ownership persist across gender, age, employment, income, and education with a respective gap of 4%, 13%, 12%, 7%, and 19% in 2021 despite narrower than the corresponding gaps of 7%, 16%, 15%, 13%, and 23% in 2017. Therefore, it is essential to have complete insight into the underlying issues, associated risks, and challenges faced by the vulnerable groups in digital financial inclusion. This understanding is crucial for bridging the gaps of inequality and for providing a more robust basis for potential policy recommendations, thereby charting a course for future research. This study conducted a systematic literature review of how previous research addressed different areas of concern and the nexus among fintech, financial inclusion, and vulnerable groups in accordance with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) protocol [11] and the protocol, search, appraisal, synthesis, analysis and reporting (PSALSAR) framework [12]. Specifically, this study investigates the antecedents, role and effectiveness of fintech in facilitating financial inclusion, thus empowering the vulnerable groups. The key issues, findings, limitations, and suggestions highlighted in the literature are systematized and summarized into themes and future research directions.

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Fig. 1 The gaps in account ownership by gender, age, employment, income, and education between 2017 and 2021 (Data sourced from the World Bank's Global Findex Database 2021)

Although there has been systematic literature review and bibliometric studies on fintech and financial inclusion, these works were either topic- or geographic-specific, focusing primarily on specific issues or certain demographic groups in certain countries or regions, which limits the generalisability of findings. None of these studies pivoted their literature review holistically on fintech, financial inclusion, and vulnerable groups. For examples, [13] conducted a systematic review on fintech services and financial inclusion of developing countries, while [254] conducted a systematic assessment and bibliometric investigation on fintech among women, and [14-15] focused on fintech and financial inclusion in rural India. On the other hand, [16-17] conducted systematic review on fintech and financial inclusion in the context of sustainable development and financial stability, respectively, whereas [18-19] did a bibliometric review on the impact and application of fintech in relation to financial inclusion, [253] synthesizes insights from existing literature on fintech-enhanced agricultural initiatives and highlights the importance of creating a robust sustainable agricultural value chains for smallholder farmers, and [255] conducted a systematic review with bibliographic mapping to identify key factors driving blockchain adoption in the banking industry. It is worth highlighting that [20] have identified five major themes from their bibliometric review on emerging trends in research in fintech and financial inclusion, including financial digital technologies and micro, small and medium-sized enterprises; fintech and digital financial inclusion; financial literacy and fintech adoption; information systems and regulation of mobile money; and financial innovation and sustainable development. [240] also highlighted four key dimensions (digital literacy/knowledge, affordability/costs, equity-deserving group-sensitive content, and availability/access to infrastructure) and barriers (accessibility, costs, digital skills, language) of digital divide among vulnerable groups. While it is essential to understand the context- and region-specific challenges faced by vulnerable groups, a holistic understanding of the associations, similarities, and distinctions of issues among these groups also offers valuable insights. This implies that research concerning the fintech-financial inclusion-vulnerable group nexus stays unexplored and has exciting potential for future studies. This study also echoes the demand for research on the fintech and

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financial inclusion of unbanked and low-income individuals and fills the gap noted by [21]. Therefore, this paper extends beyond the limited scope of previous review studies, offering deeper insight into the relationship between fintech, financial inclusion, and vulnerable groups. The comprehensive review of relevant literature from a broader perspective encompassing all categories of vulnerable groups enables this study to identify potential gaps in current studies and opportunities for future research. This study also plugs the knowledge gap between fintech and vulnerable groups and may serve as a reference for governments in devising policies and strategies to accelerate digital financial inclusion, and for fintech companies and financial institutions designing products and services targeting vulnerable groups. The outline of the paper is as follows. Section 1 provides a brief introduction to the research topic, followed by an overview of fintech, financial inclusion, vulnerable groups and theoretical frameworks used in fintech adoption research in Section 2. Section 3 presents the methodology and Section 4 reports and discusses the results. The future research directions are proposed in Section 5 and lastly, the paper concludes in Section 6.

2. An overview of fintech, financial inclusion and vulnerable groups

Fintech can be regarded as “a new financial industry that applies technology to improve financial activities.” [22], or more specifically, “a technology-enabled innovation in financial services that could result in new business models, applications, processes or products with an associated material effect on the provision of financial services.” [23]. The body of fintech literature has predominantly emerged after 2015, and numerous review articles have been published in the field since then [24-27]. Fintech innovatively uses technologies such as the Internet, mobile applications, big data mining, blockchain, and artificial intelligence (AI) to improve the processes and applications of financial products and services [28]. Fintech has significantly reformed the traditional business models of financial institutions and businesses, and the way how banking and financial transactions were conducted, that is, from physical platform, hardcopy documents, and human advisor to online, softcopy, and robo-advisor, thereby reaching out to more people and making financial services more accessible, efficient and affordable [29-31]. Examples of fintech innovations include online savings, payments and investments, digital insurance and cryptocurrencies, and robo-advisors, crowdfunding, and P2P lending platforms [32-33].

Fintech has been shown to facilitate financial inclusion and SDGs [3-4, 13-14, 17]. Specifically, fintech accelerates financial inclusion by expanding: “access to appropriate, low cost, fair and safe financial products and services from mainstream service providers” [34] and termed as: “a saviour of economies from poverty, crises, and underdevelopment” [13]. Mobile money, for example, has facilitated financial inclusion and enabled more equal access and account ownership opportunities, in which mobile money accounts has markedly increased from 5% to 15% from 2017 to 2024, and reduced the gender and income gaps in account ownerships in developing economies to 4% and 11% in 2024 from 9% and 14% in 2017, respectively [10]. These same facts were echoed by the finding on the fintech services and financial inclusion of developing countries, which indicated that fintech has expedited financial inclusion through mobile payment and lending services [13]. The review paper also pointed out that fintech services and financial inclusion are important for vulnerable populations (e.g. women, youth, and rural communities) and contribute to the reduction in inequality and income disparities between genders and urban-rural areas.

Despite the encouraging progress over the past decade, a few challenges remain in achieving greater financial inclusion, particularly among the vulnerable groups and low-income countries, due to geographical constraints (lack of infrastructure and long distances), unaffordability and high costs (high transaction costs and

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interest rates), inappropriate products and services (fail to meet the needs of a particular demographic), and financial illiteracy (lack of financial awareness and knowledge) [14]. Furthermore, not all the literature unanimously agreed that financial inclusion unconditionally supports country development and poverty reduction. Although these studies generally concur the positive impact of financial inclusion, the effect varies between regulatory frameworks and geographic regions, subject to, for examples, inclusive mechanisms, financial and economic conditions, and the levels of poverty, financial innovation, and financial literacy [17, 35-36]. Therefore, in light of the aforementioned literature and the future research directions recommended by [13], that is, to focus on the interrelation between fintech and financial inclusion across various socio-economic dimensions and to explore the behavioural patterns of fintech adoption and usage among women, youth, micro-entrepreneurs, rural and low-income communities, this paper attempts to offer a comprehensive review of literature involving fintech, financial inclusion and vulnerable groups. It seeks to enhance understanding of the relationships and reconcile the conflicting views on the impacts of fintech on financial inclusion and vulnerable groups.

3. Methodology

This paper refers to the PRISMA protocol [11] and PSALSAR framework [12], as illustrated in Figure 2 and Table 1, to systematically search, screen and review published research articles indexed in the Scopus database, focusing on fintech, financial inclusion, and vulnerable groups. The PRISMA and PSALSAR are widely used by scholars for systematic reviews as they provide guidelines, flow diagram and checklist to ensure the transparency, consistency and high standards of searching, evaluating, reviewing and reporting protocols. Scopus database is considered in this study as it is accessible and is among the largest and most frequently cited scholarly databases with a collection of more than 97.3 million records and 28.3 thousand active serial titles, in which nearly 99.11% of the collection is similar to the serial titles indexed in the Web of Science database and it is accessible in this study [37-38]. This study covers English-language, peer-reviewed journal articles of all types and research designs, published before 2025 in the multidisciplinary and all social sciences subject areas, including accounting, business and management, economics and econometrics, finance, decision sciences, arts and humanities, psychology and social sciences. These articles were initially searched using specific keywords (as shown in Figure 2) that were identified from review of literature and definition of vulnerable groups [1-2, 13, 16]. An initial sample of 827 articles that met the keywords and search criteria in the ‘title’, ‘abstract’ and ‘keywords’ fields of articles was shortlisted. Next, a total of 576 articles not entirely related to fintech, financial inclusion, and vulnerable groups were excluded after the initial screening based on the articles’ titles and abstracts, resulting in a total of 251 articles remaining. Additional screenings of eligible records limiting to accessible articles followed by main body skim-reading of articles further excluded 32 and 34 articles, respectively. This has resulted in a final sample of 185 articles focusing exclusively on fintech and financial inclusion of vulnerable groups. The articles’ information (e.g., title, years, sources, subject areas, authors, affiliations, countries, citations, and funders) was downloaded from the Scopus database, and the details of each article (e.g., vulnerable group category, research objectives, research methodology, sampling methods and analysis models, main findings, limitations, suggestions for future research, and conclusions) were entered into a spreadsheet for content analysis.

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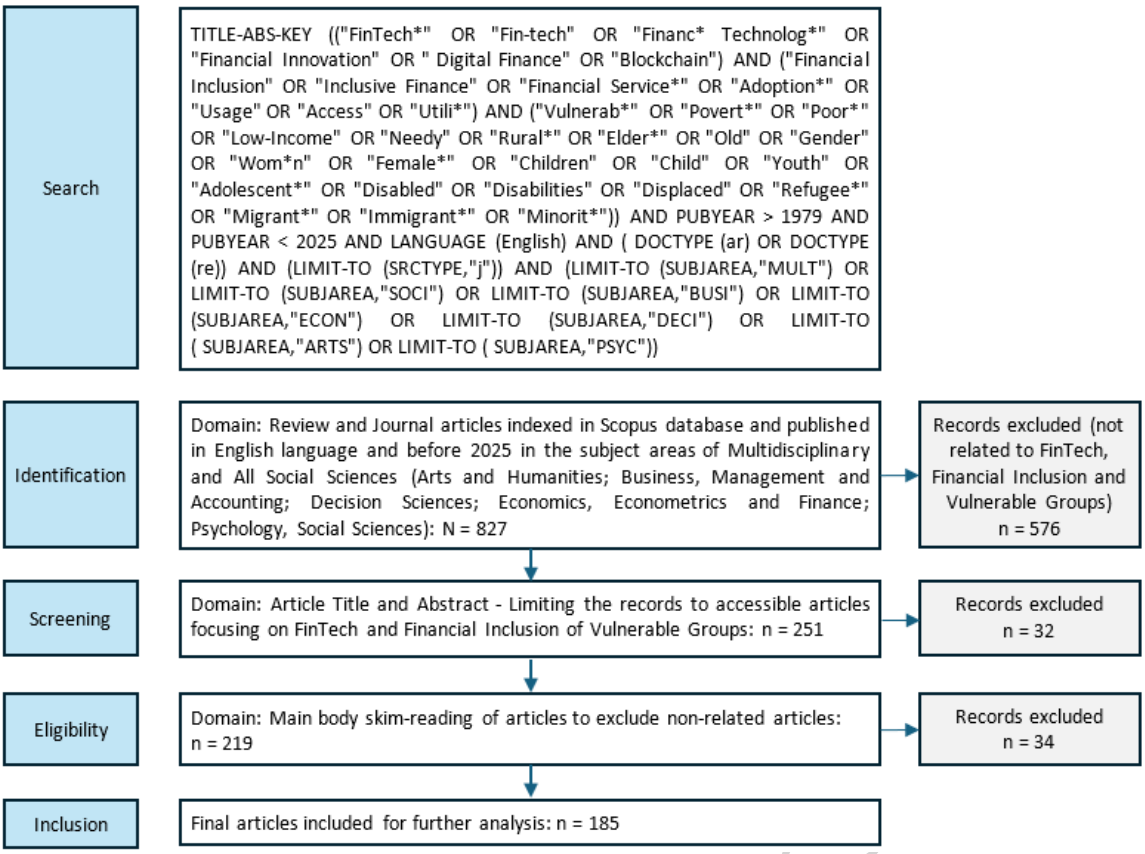

Fig. 2 PRISMA protocol for systematic literature review screening process [11]

Table 1 The PSALSAR Framework [12]

PSALSAR Stage	Methods
Protocol	Research scope: Systematically review journal articles concerning fintech and financial inclusion of vulnerable groups.
Search	Search strategy: Related keywords "(FinTech OR Fin-Tech OR Finance Technology OR Finance Technologies OR Financial Technology OR Financial Technologies OR Financial Innovation OR Digital Finance OR Blockchain) AND (Financial Inclusion OR Inclusive Finance OR Financial Service OR Financial Services OR Adoption OR Usage OR Access OR Utilise OR Utilize OR Utilisation OR Utilization) AND (Vulnerable OR Vulnerability OR Vulnerabilities OR Poverty OR Poverties OR Poor OR Poorest OR Low-Income OR Needy OR Rural OR Elderly OR Elderlies OR Old OR Gender OR Woman OR Women OR Female OR Child OR Children OR Youth OR Adolescent OR Disabled OR Disabilities OR Displaced OR Refugee OR Refugees OR Migrant OR Migrants OR Immigrant OR Immigrants OR Minority OR Minorities)"
Appraisal	Method of selection of articles using PRISMA protocol.
	Inclusion criteria: English-language peer-reviewed research or review articles published before 2025 in the subject areas of multidisciplinary and all social sciences.
	Exclusion criteria: Any articles that are inaccessible and not related to fintech, financial inclusion and vulnerable groups.
Synthesis	Records extraction and categorization using Excel and PRISMA checklist.
Analysis	Analysis, results and discussion: Descriptive and content analysis, classification of literature, findings and discussion, conclusion and future research directions.
Reporting	As a systematic literature review journal article.

4. Results

4.1 Descriptive summary and theoretical frameworks

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The descriptive summary and the number of articles and citations of research in fintech, financial inclusion, and vulnerable groups are presented in Table 2 and Figure 3, respectively. Table 2 shows a total of 185 peer-reviewed journal articles published in 136 indexed journals between 2017 and 2024, involving 160 authors and affiliations from 64 countries, in which 46.5% (or 86 out of 185) of the works were funded by various agencies and organizations. The Peking University, Shandong University of Finance and Economics, and University of International Business and Economics led the top three authors' affiliations, while the top three authors' countries are China, United States, and United Kingdom. This may be partly contributed by the greater data accessibility, strong research ecosystems and funding support, as evidence by the top three listed funders, along with the relatively advanced fintech and regulatory frameworks in these countries [16, 241]. The classification of the articles by vulnerable group shows that research concerning fintech, financial inclusion, women, and poverty has received the most attention with 46 articles each, followed by research related to rural communities (41). In contrast, research involving people with disabilities and the elderly have received the least scholarly attention, with only 5 and 6 articles published, respectively, prior to 2025, which may be due to limitations in data availability, research priorities, or their underrepresentation in fintech usage. As for the classification of the articles by research method, Table 2 shows that nearly 75.14% (or 139 out of 185) of the articles involved quantitative data analysis, in which 86 articles used secondary data, while 53 articles involved primary data collected via questionnaires. There are 7 qualitative, 5 descriptive and 4 mixed-method research, while the remaining 30 articles are experimental, review, conceptual, case study, discussion, review, and bibliometric studies.

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Table 2: Descriptive summary

		f			*f*
Article information	Year range	2017-2024	Article information	No. of authors	160
	No. of articles	185		No. of affiliations	160
	No. of sources	136		No. of countries	64
	Average no. of citations per article	27.29		No. of funders	86
Top 3 affiliations	Peking University.	6	Top 3 countries	China	57
	Shandong University of Finance and Economics.	6		United States	24
	University of International Business and Economics.	4		United Kingdom	24
Top 3 sources	Sage Open	6	Top 3 subject areas	Economics, Econometrics and Finance.	98
	China Economic Review	5		Business, Management and Accounting.	72
	Sustainability Switzerland	5		Social Sciences.	72
Top 3 average citations per year	[39] Demir et al. (2022)	111.3	Top 3 funders	National Natural Science Foundation of China.	16
	[40] Gabor and Brooks (2017)	57.3		National Office for Philosophy and Social Sciences.	15
	[41] Mhlanga (2020)	55.0		Fundamental Research Funds for the Central Universities.	9
Category of articles by vulnerable group	Children and young people	10	Category of articles by data and	Descriptive	5
	People with disabilities	5		Mixed-Method	4
	Elderly	6		Qualitative	7
	General	22		Quantitative (Primary Data)	53

Migrants and minorities	9	research method	Quantitative (Secondary Data)	86
Poverty	46		Others (review, conceptual, case study, discussion, experimental and bibliometric)	30
Rural	41			
Women	46			

In addition, Figure 3 shows that the total and average citations of 185 articles from 2017 to 2024 are 5048 and 27.29 per article, respectively. There has been an apparent increasing trend of publications and citations of articles concerning fintech, financial inclusion, and vulnerable groups since 2017. A marked and significant increase in the numbers was observed, particularly between 2021-2024, in which the number of publications and citations have increased more than fivefold and eightfold respectively over three years, rising from 12 publications and 260 citations in 2021 to 65 publications and 2,138 citations in 2024. Specifically, there has been a notable rise in citations related to gender, fintech and AI, China and Africa, indicating growing interest in these areas. The upward trends in publications and citations echo the trends reported in fintech, financial inclusion, and SDGs related scholarly publications [16, 42-45], and also indicate that research interest in the fields of fintech, financial inclusion, vulnerable groups, and SDGs have received tremendous attention from scholars in the past few years, especially when the target date of the World Bank to achieve the SDGs by 2030 is approaching [6].

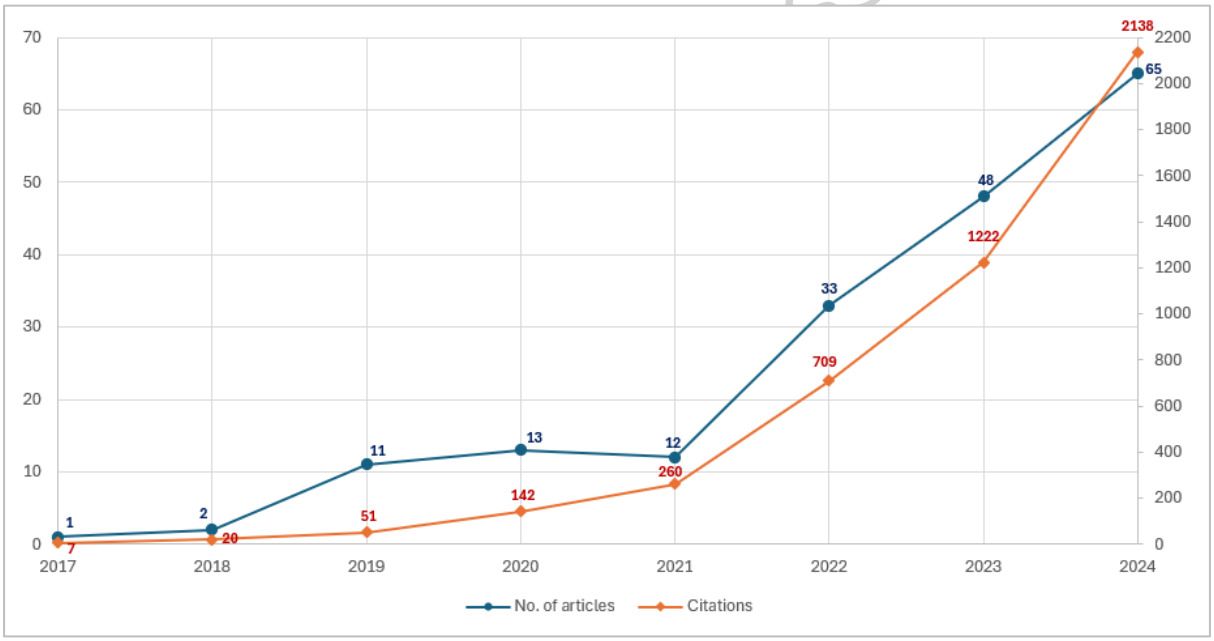

Fig. 3 Number of articles and citations of articles concerning fintech, financial inclusion and vulnerable groups (2017-2024)

The summary of theoretical frameworks used in fintech adoption research among vulnerable groups is presented in Appendix A. Based on the review, the most widely used theories are the Technology Acceptance Model (TAM), followed by Theory of Planned Behaviour (TPB), the Unified theory of acceptance and use of technology (UTAUT), and its extended version, the UTAUT2. For example, studies [142-143] and [235] employed TAM to examine fintech adoption among women in Indonesia and India, respectively. Similarly, [187] and [189] focused, respectively, on fintech adoption among semi-urban university students in Ghana and college students in India. Moreover, [181], [190] and [216] used TAM to investigate the determinants of mobile wallet among rural millennials in Oman, young people in

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Malaysia, and digital immigrants in Indonesia, respectively. [194], on the other hand, explored the determinants of P2P lending intention among youths in Indonesia. A few researchers have integrated TAM with other theories to gain a better understanding of the antecedents of fintech adoption. For example, [204] integrated TAM with the Social Influence Theory to examine the effects of various social dynamics and digital literacy on elderly engagement with mobile banking in China. Likewise, [236] combined TAM with the Diffusion of Innovation Theory, and Consumer Behavioural Theory to identify the factors influencing fintech adoption among Indian women, and [188] merged TAM with TPB to study the determinants of young people's intention in Vietnam to continue using digital financial services. The TPB has also been used to investigate women's adoption of fintech and cryptocurrencies in Bangladesh and Spain, respectively [122, 134]. Also, [141] applied TPB to examine the effect of digital financial literacy on financial decision-making among women in India. Researchers have also combined TPB with other theories, for example, the Antecedent-Behaviour-Consequence Model, and Gender Schema Theory were used to examine the continuous use of fintech of vulnerable groups in Ghana [63]; the Bottom of Pyramid Theory was employed to examine the effect of inclusive finance knowledge on Chinese farmers' financing intentions [186], and the Social Role Theory was used to examine the impact of digital financial literacy on cashless payment systems among Jordanian university students [193]. The UTAUT and UTAUT2 models are also widely used as guiding frameworks in fintech adoption research. Research by [128], [178], and [191] employed UTAUT to investigate the impact of financial literacy on Indonesian women's fintech use and business success; the factors and effects of digital finance platform usage on personal finance management among rural populations in China; and the determinants of fintech service adoption intentions among business students in Brazil, respectively. In contrast, [139] integrated UTAUT with Social Cognitive Theory to identify the determinants of ZakaTech adoption in Saudi Arabia. Alternatively, UTAUT2 has been applied to understand the disposition of low-income Colombians using automated banking services [77]; to identify the determinants of mobile banking for zakat distribution among *asnaf* in Malaysia [86], and to investigate sustainable digital financial inclusion among Indian women in banking and IT sectors [234]. Additionally, researchers have also merged UTAUT2 with the Appraisal Tendency Framework to examine the impact of mobile money on the access and the use of financial services in Uganda during the COVID-19 pandemic [95].

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4.2 Fintech, financial inclusion and vulnerable groups

The present review shows that despite fintech can facilitate financial inclusion, challenges persist especially in providing equal and easy access to financial products and services, and eradicating inequalities among different geographic, demographic, and socio-economic groups. For instances, although [46] demonstrated that mobile payment usage could increase happiness of the elderly, less educated, and rural communities in China, [47, 48] both found that, in India, the adoption of fintech varies across demographic, social, and economic status, where women, the elderly, and less educated people were found to be relatively less involved in digital transactions than their counterparts despite tremendous progress in financial inclusion in the country. [49] also pointed out that Argentines living in marginalized areas with lower education and income face significant barriers in accessing digital financial and credit services. Similarly, [50] documented that fintech does not contribute much to financial inclusion of the elderly, women, less educated and lower social class communities in the Middle East and North Africa (MENA) region. [51] also found in Saudi Arabia that being male, wealthy, and educated have relative advantages in financial inclusion, but in contrast, high costs, insufficient supporting documents, accessibility and reachability of financial services are main barriers to financial inclusion. On the other hand, [52] highlighted that although fintech resolves the geographic barrier

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and increases traditional financial products usage in Kenya by nearly 4%, it could not effectively address financial inclusion inequalities of women, the less educated, poor and young Kenyan people, may be due to traditional marginalization or high costs of technological devices. This is corroborated by [53], who showed that the same vulnerable groups were among those who are most financially excluded in the Western African Economic and Monetary Union region. Among the major obstacles to financial inclusion in the region are lack of resources, trust, and religion, in addition to excessive costs and documentation. Besides, the pace of the development of digitalization and financial inclusion through fintech is also influenced by emotional connections with money and affected by security, information privacy, and fraud issues [54]. Individuals often associate tangible money with control, security and accountability, in which high transaction costs and perceived risks may deter fintech adoption, especially among financially illiterate and vulnerable individuals, due to issues of intangibility, transparency and trustworthiness in fintech. Likewise, security, information privacy and fraud concerns may hinder fintech adoption as fintech processes vast amounts of sensitive personal data and thus any fraudulent activities and data breaches would undermine confidence in fintech.

The extent and coverage of financial inclusion also depend on the advancement of technology such as blockchain, AI, and the adoption of fintech products and services. More specifically, [55] demonstrated that blockchain can improve the transparency and accountability of contested markets and thus made contested markets accessible to vulnerable groups, which in turn, lend support to social and economic systems. Meanwhile, [41] found that AI can promote digital financial inclusion by providing support in, for example, customer support and helpdesk services using chatbots, information asymmetry reduction, and automated fraud and risk alerting systems. This is in line with findings indicating that digital financial inclusion leads to financial inclusion of those who were previously excluded, such as those from informal sectors, youth, unemployed, and low-income populations in Africa [56], and facilitates entrepreneurship in East African, especially among female, young, and rural mobile money users in Kenya and Uganda [57]. Similar finding was found in China where mobile banking applications promote financial inclusion of small and micro enterprises, the rural and poor households through loans from village and township banks [58]. [59] also observed a positive association between P2P lending and both digital and financial inclusion in the U.S., particularly in counties with more vulnerable populations and limited commercial bank loans. Research also found a positive interaction between biometric identification technologies, mobile money, digital literacy, and digital financial inclusion, which collectively explained 29% of the variations in digital financial inclusion of Ugandan unbanked vulnerable populations and people with disabilities [60].

Additionally, research also suggested that fintech adoption and financial inclusion were subjected to various demographic factors such as gender, age, occupation, education, income, and marital status [61]. [62] documented that age has an inverted U-shaped effect on the financial inclusion of the Association of Southeast Asian Nations (ASEANs) countries, in which a positive and negative relationship occurring before and after the ages of 29 and 45 years old, respectively. [63] also found that the intention and actual use of mobile payments over time in Ghana were moderated by gender, age, and the education level. In a different study, [64] showed that higher financial literacy enhances digital finance services usage such as online payments and borrowing in China, particularly among the elderly, low-income and rural families. This is consistent with the finding of [65], who showed that digital and financial literacy are among the main antecedents of financial inclusion and resilience in South Asia and Sub-Saharan Africa (SSA) regions, and heterogeneities were observed across regions, and among females, the poor, and rural households. The same in the MENA region, [50] found that educational attainment and employment status, are among the most important antecedents

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of digital financial inclusion. The finding implies the need to redefine traditional financial literacy to include digital literacy to improve household financial resilience and financial inclusion in the long term. Among the suggestions put forward to increase financial inclusion are improved digital infrastructure and financial literacy education for the rural and marginalized communities [47], expand the application of AI in financial products and services to ease and encourage active participation of the vulnerable groups in formal financial market [41], and implementation of digital financial service taxes, at best with minimal adverse impacts on vulnerable groups and fintech users, to support governments expenditures particularly in developing new technological infrastructures and policies considering that the realization of fintech adoption, financial inclusion, and SDGs in the long-term also relied on taxation [56].

In demographic contexts, studies show that countries with high gender discrimination face lower financial inclusion, where women often exhibit lower financial literacy and trust in fintech, which hinders fintech adoption [110-111]. However, fintech has enabled women's entrepreneurship and lead to greater financial inclusion and thus improve their bargaining power at home and narrow wage gaps in countries like China, Indonesia and Kenya [115-116, 128, 133]. Fintech has also shown potential in reducing poverty and income inequality among the poor and rural communities, however, their financial inclusion, is often constrained by financial literacy, and limited access to digital infrastructure, such as smartphones and internet, as well as high cost of digital financial services, which are often unaffordable for low-income communities [78-80]. Despite these barriers, fintech has demonstrated positive impacts such as improved savings behaviour, entrepreneurship, access to credit, and financial security among rural and low-income earners, particularly in China, Africa and Bangladesh [56, 75-76, 81, 84, 165-166]. The elderly also face significant barriers to financial inclusion due to digital literacy and perceived complexity of fintech products further exacerbate exclusion [64, 204]. Perceived ease of use, perceived usefulness, trust, and digital use experiences are found to be among the key factors that influence the adoption of mobile banking platforms among the elderly in China [203, 216]. The ethnic minorities and migrants continue to face persistent barriers to financial inclusion. These barriers are often rooted in systemic inequalities and socio-economic marginalization, including limited access to formal identification, language barriers, and discriminatory practices in financial systems [214]. However, targeted interventions, such as mobile banking training for migrant families in Bangladesh, have shown promising improvement in fintech adoption and financial decision-making [209]. People with disabilities also face both social and technical barriers to financial inclusion, including inconvenient access to digital platforms, and lack of assistive technologies [197-198]. Despite these challenges, fintech has been shown to improved financial access and well-being of some individuals with disabilities by reducing mobility-related barriers and simplifying transactions [199-200].

Overall, the literature on the one hand, showed that fintech can accelerate digital financial inclusion, mitigate inequalities, and improve the wellbeing and financial sustainability of vulnerable groups conditional on factors such as geographic [49], demographic [52, 63], financial literacy [64], and digital infrastructure development [59]; on the other hand, it was revealed that the relatively disadvantaged groups such as women, the less educated, low-income, and rural populations, to some extent, still face challenges in financial inclusion [48-49, 57]. However, the literature also suggests that financial inclusion can be more expansive, particularly to vulnerable groups and in rural areas by removing barriers through increased awareness, digital security and trust in financial institutions, minimal documentation and transaction costs, improved financial and technological infrastructure, reduced distance to access, use of quality financial products and services [51-53, 63, 65-66], and more entrepreneurial activities [46, 57]. Another major barrier to the financial inclusion of the vulnerable group that

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requires special attention is the low level of education and digital financial literacy. As pointed out by [49], the financially and credit-excluded population, who face greater barriers in accessing digital financial services, typically has the characteristics of being less educated, having lower and less stable income, and living in marginalized areas. These vulnerable groups are usually exposed to the risk of financial exclusion, probably because they generally have low educational attainments and limited knowledge of fintech and finance. This may explain why digital finance inclusion adversely impacted the life satisfaction of Chinese elderly people [67], and why fintech was not conducive to MENA's financial inclusion, particularly among women, the less educated and elderly people [50]. This is in line with findings showing that education is an essential factor in reducing the barriers to financial inclusion [51], the most important determinant of fintech adoption [48] and has a positive impact on financial inclusion [68]. Financial literacy is important in promoting digital financial services adoption, especially for more complex financial products and among vulnerable groups [64]. This is also corroborated by finding that demonstrated digital literacy significantly positively moderates the effect of fintech on digital financial inclusion in rural areas in Uganda, particularly among the vulnerable populations and people with disabilities [60]. Lastly, the literature showed that fintech does not necessarily advance financial inclusion of vulnerable groups if the obstacles and antecedents of accessibility of financial products and services have not been properly addressed. Although research has shown that countries with low gender gaps tend to have lower financial inclusion for women [111], and most vulnerable groups often face common barriers to fintech adoption, such as digital financial literacy, trust, affordability and accessibility, the unique challenges faced by different vulnerable groups also deserve close attention. For example, women are often hindered by low empowerment and restrictive cultural norms; low-income individuals are frequently excluded from formal credit systems; rural populations are typically geographically isolated; and the elderly and people with disabilities are frequently deterred by the absence of supportive and assistive technologies. Therefore, developing feasible and effective policies with specifically tailored initiatives in promoting digital financial inclusion and education, helping vulnerable groups in using sophisticated digital financial products with the necessary digital financial knowledge, trust, and confidence are vital in achieving equitable financial inclusion and reducing inequalities within and across countries in the current digital age. The list of relevant references in fintech, financial inclusion, and general vulnerable groups is presented in Appendix B.

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4.3 Fintech, financial inclusion and poverty

Research related to fintech, financial inclusion, and poverty (see Appendix C) have attracted much attention from scholars, considering that fintech and financial inclusion have been promoted as one of the key pillars for achieving the United Nations' no poverty (SDG 1) and reduced inequalities (SDG 10) sustainable development goals [6, 69]. Among the scholars are [39], who used a panel data of 140 countries to investigate the connection between fintech, financial inclusion, and income inequality, and [70], who used data from 119 countries to examine the association between the income inequality and accessibility of financial services. Furthermore, quantile regression analysis of [39] revealed that fintech, through financial inclusion, reduces income inequality at all quantile levels, while [70] found that financial services accessibility reduces income inequality in countries with less development, but the association is weaker in countries with more fintech, implying that fintech is a substitute for traditional banking services. Similarly, from a panel data of 25 African countries, [71] showed that fintech, mediated by financial inclusion, played a fundamental role in the reduction of income inequality. In other related studies, both [72-73] examined the effect of both fintech and financial inclusion on poverty, in the context of selected ASEAN countries. In their study, [72] revealed that both fintech and financial inclusion

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increased the GDP but reduced the unemployment and poverty rates, while [73] found that financial inclusion significantly reduced poverty. In general, the literature showed that fintech improves financial inclusion and thus reduces poverty, particularly in developing countries, through improved infrastructure, accessibility, financial literacy, and institutional supports. The finding is consistent with [242-243], which found that fintech and financial inclusion could significantly alleviate poverty in developing countries, in contrast to high-income countries, by reducing gender, infrastructure and access gaps. In other words, governments play a crucial role in mitigating poverty by establishing regulatory frameworks and digital infrastructure that support fintech companies and promote financial inclusion for vulnerable groups in developing countries. Inclusive financial policies that ensure data privacy, provide cybersecurity protection, and enhance digital financial literacy are equally important [244]. Another group of researchers focused on country-level context of fintech, financial inclusion, and poverty. For examples, [74] found in the U.S. that fintech enhanced the financial services accessibility for citizens and businesses in regions with fewer traditional banking systems, lower incomes, and more minority households, while [75] examined the effect of mobile financial services on economic opportunities and welfare of low-income textile factory workers in Bangladesh. They found that financial innovation to some extent improved fund transaction speed and security, and the savings of low-income earners. On the other hand, [76] examined whether the financial inclusion of the urban poor in Kenya has improved following the introduction of mobile money in the country. The result showed that the Kenyan urban poor were able to improve their savings through the adoption of mobile money, which better suited their needs and significantly improved their financial inclusion. Another evidence supporting the role of fintech in driving financial inclusion is provided by [77], who demonstrated that technology disposition is closely associated with automated banking services usage, thus indicating the potential of fintech in driving financial inclusion of the poor in Columbia. Similarly, relevant studies in the Chinese context also found fintech and digital financial inclusion can mitigate income inequality and alleviate poverty, particularly in low-income regions and among households with high levels of education, digital tools, dependency ratios [78-80]. However, the reviewed literature also revealed that research on the long-term effects of fintech on poverty remains limited. For example, among the few existing studies on the topic, [245] found that mobile money, in the long run, increased financial resilience and savings of Kenyan households headed by women, thereby reducing extreme poverty in Kenya. Similarly, a review by [243] discovered that digital technologies contribute to long-term poverty alleviation subject to improved financial literacy, entrepreneurship and asset accumulation.

Fintech and financial inclusion were also shown to be associated with entrepreneurial behaviours and Islamic finance. Research found that fintech can lower formal credit constraints, enhance risky financial assets allocation, and promote entrepreneurial behaviours, thereby alleviating the possibility of relative poverty of rural households [81]. [82] and [83] also showed that digital financial inclusion, mediated by entrepreneurial activities, can alleviate rural household vulnerability to poverty in China, particularly in Chinese households with low-income, and rural areas with less financial development and human capital. Additionally, [84] investigated fintech's role in encouraging savings for microenterprises in 74 low- and middle-income countries, arguing that the access to banks moderates the effect of fintech adoption on savings for poor women microentrepreneurs. More specifically, fintech is more effective with bank accounts, and the complementarity between fintech and banks is more impactful for poor women. From a sample of 81,345 individuals, the result shows a positive correlation between mobile money and savings for micro-enterprises, particularly among the poor, women, and individuals with bank accounts. The finding suggests that complementarity exists between banks and fintech, indicating the equal importance of inclusivity in formal banking institutions and fintech in alleviating poverty. The finding offers an alternative to the popular discussion that fintech is

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a substitute for banks and suggests the need to continue driving financial inclusion with formal banking. As for research in Islamic fintech, [85] used a qualitative approach and showed that Qard Hasan has the potential of becoming an Islamic crowdfunding instrument in financing micro-enterprises, thereby alleviating poverty in Malaysia, with the condition that its associated risks (e.g. default and funding risks) are manageable. Meanwhile, [86] offers an interesting case study on the factors of mobile banking usage among the *asnaf* (a group of people that is eligible to receive zakat aid or Muslim tithe) in the Malaysian context by using UTAUT2 as a guiding framework, with additional constructs such as religiosity, attitude and trust (initial trust model). The result from a sample of 470 *asnaf*, and the structural equation modelling approach reveals that *asnaf*'s intention to use zakat payments via mobile banking is conditional on factors such as their religiosity, trust and habit, as well as the performance expectancy, effort expectancy, and facilitating conditions of using mobile banking. The finding can be a reference for the Malaysian government to promote fintech adoption and to improve the efficiency of zakat distribution by the zakat institutions in the country. This research adds to the Islamic fintech literature and Islamic countries by addressing Islamic social financing and has social implications to the Muslim society's poverty alleviation. Researchers have also identified other potential mechanisms and conditions, offering several suggestions for mitigating inequality and poverty issues through greater financial inclusion. Among the recommended mechanisms and solutions are information and communication technologies (ICT) and pro-poor strategies. [87] analysed panel data on 62 countries and showed that ICT diffusion could expand financial inclusion and hence reduce poverty and inequality, accelerating economic growth, whereas [88] conducted a case study on three fintech organizations to study the innovative strategies adopted in addressing liability of poorness and promoting entrepreneurship among the poor in South Asia. They proposed an emerging typology of fintech, highlighting pro-poor strategies such as simplified procedures, and high interactions among digital, digital-human and human. In another research, from the mixed-method analysis result that showed low-income parents actively used trusted and convenient digital financial services than those provided by traditional banks, [89] concluded that low-income parents typically consider the practicality of financial services and are actively seeking out services on various platforms that could be trusted for their transparent fees and ability to meet their needs. [90-91] in contrast, ascertained the views that improved digital technology, digital literacy and skills, and business regulations are important drivers for the fintech use and financial inclusion of low-income communities. This is echoed by [92], who suggested that specific laws and policies should be enacted to encourage the use of digital financial services and currencies among the low-income populations. [93], on the other hand pointed out that as bank branch expansion remains relevant and facilitates fintech usage and financial inclusion, a hasty move to digital transformation could delay the improvement of low-income rural household welfare and advancement of shared prosperity in China. Additionally, the adoption of fintech was also influenced by the uncertain and stressful situations such as the COVID-19 pandemic. [94] found that fear, social influence, awareness, and trust stimulated the adoption of fintech services among low-income segment customers in India during the pandemic. [95] also found that access and usage of financial services in Uganda were accelerated during the pandemic via contactless fintech products such as mobile money.

However, not all studies show consistent evidence of a positive linear association between fintech, financial inclusion and poverty. A strand of research found that fintech has a nonlinear relationship with income equality and poverty. Specifically, [96] used data from 45 SSA countries and observed a U-shaped association between fintech and poverty, where fintech development initially reduces the rate of extreme poverty, but then the effect starts to decrease after reaching a certain threshold of fintech penetration. This may be because fintech adoption initially enhances financial inclusion and thereby reduces poverty; however, excessive or

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unregulated fintech implementation may later increase the risks of over-indebtedness and financial vulnerability. In contrast, [97] observed that digital finance and household income inequality are inverted U-shaped in China. This implies that as fintech develops, income inequality initially widens before eventually narrowing or in other words, poverty first reduces but increases later after a certain point. This may be caused by an imbalance in digital infrastructure, financial literacy and economic opportunities between urban and rural areas at the beginning, but the development of pre-requisite infrastructure, financial education and fintech gradually enable low-income households to have more access to financial services and entrepreneurial opportunities, helping them improve their overall income and thereby reducing the income inequality. In another strand of literature, [36] conducted a critical policy discourse analysis on pro-poor intervention using digital finance and found insufficient evidence of digital finance promoting socioeconomic development of poor people. Researchers such as [71, 73] also found that financial innovation and exclusion widen income inequality, possibly because of market imperfections and information asymmetries that adversely affect the low-income communities who are typically less digitally literate and have limited digital devices and internet access. Furthermore, [98-99] argued that digital financial inclusion in Kenya and Ghana, respectively, did not benefit the poor and end poverty, but instead has resulted in the exploitation of working-class labour with indebtedness from digital microloans, high transaction costs, excessive taxation, and prevalence of dormant accounts. Additionally, in review conducted by [100] on the impact of digital financialization on micro-entrepreneurs also showed that micro-entrepreneurs in India might have difficulty adapting or have little incentive to use new financial technologies, and therefore remained heavily dependent on indigenous moneylenders, resulting in a high cost of subprime financing. These findings were substantiated by studies of [101], which found a persistent rise in bankruptcies among the poor impacted by sharp declines in access to marketplace lending and new lending technology. Similarly, [74] observed that fintech is disproportionately served in the U.S. even though it generally improves financial inclusion.

In summary, a large volume of literature has shown that fintech and financial inclusion could alleviate vulnerability to poverty and income inequality, particularly among the low-income and rural communities [39, 70, 78, 82-83, 87, 102], which can be achieved through infrastructure development [103], entrepreneurship, lowered credit barriers [81], and fintech as it enabled frequent saving in small amount [76], and reduced transaction costs and information asymmetry [80]. Researchers have also pointed out that fintech per se may not be sufficiently effective in addressing poverty issues [84]. Support from the governments and other financial institutions, as well as augmentation of other mechanisms are necessary to alleviate poverty among the vulnerable groups. For example, [104] suggested that both fintech and microfinance are necessary to reduce poverty, while [75] highlighted that in addition to fintech, parallel improvements in technology and financial education are also important. Furthermore, [84] demonstrated that poverty can be significantly reduced by mobile money combining with bank accounts, and [93] emphasized the importance of bank branches along with fintech in poor and rural areas. A proper business regulatory framework that aligns with fintech adoption is also important to drive financial inclusion and socioeconomic development in low-income economies [91]. It is also worth highlighting that although most of the literature holds the view that fintech and financial inclusion mitigate income inequality and poverty, the benefits come with associated risks and concerns. For example, [98] pointed out that the benefits of digital finance not only did not reach the poor, but it also shift the risks to them. The same concern was also raised by [105], where apart from improved financial management, the poor gain no benefit from transition of microfinance to digital finance but instead would expose them to risks of financial system. Another study also showed that although fintech provides extra employment and entrepreneurial opportunities for the migrant and

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unemployed populations, it brings little benefit to the elderly, disabled, and laid-off workers [106]. Other concerns include the high costs of digital finance services, and products and services that best meet the needs of the poor. [107] found high mobile money charges in the SSA region due to tax imposition policies and influx of foreign-owned mobile money companies into the region, and thereby causing the poor being excluded from the service. [89] also showed that low-income mothers are more interested in trustworthy financial services that have transparent fees and ability to meet their needs. In short, the literature has shown the growing importance of fintech as a market-based facilitator to financial inclusion and a potential solution to inequality and poverty issues. However, to expedite digital financial inclusion, especially among the low-income groups, more in-depth empirical research on financial inclusion and financialization processes and impacts [40], a tax policy that does not disproportionately burdens the poor [99], and practical fintech products and services with reasonable fees [69] are most needed.

4.4 Fintech, financial inclusion and women

The topics concerning fintech, financial inclusion, and women have also received great attention from the research community (see Appendix D), echoing the call for promoting women's empowerment to achieve gender equality, aligned with UN SDG 5 (gender equality) goal [6]. Literature shown that women's financial empowerment, fintech, and financial inclusion are positively associated [108-109]; however, as shown by [110] using data from 113 countries, this relationship is found to be significant only in countries with less gender discrimination, implying that gender inequality may be one of the barriers that hinder financial empowerment of women. This is substantiated by the finding of [111], who evaluated the performance of 109 countries in promoting digital services and financial inclusion among women and observed that countries with higher gender inequality have greater financial exclusion for women. Similar findings were found in country-level data. For example, [112] examined gender gap in financial inclusion and reported that Pakistani women have lower account ownership, credit and debit card usage, and utilization of digital financial services compared to men. In Nigeria, [113] analysed the spatial and disaggregated data and found evidence of socioeconomic inequalities, with gender being identified as one of the main influencing factors of geographic clustering of financial inclusion. The financial inclusion in Jordan is also found to be different between genders attributed to differences in demographic characteristics such as wealth, employment, education, and income levels [114]. Likewise, in China, [115] demonstrated that fintech narrows wage gap between genders, and promotes Chinese women's entrepreneurship and empowerment, while [116] provide empirical evidence that the use of digital finance could significantly improve Chinese women's empowerment at home, but vary among households from different regions, living patterns, and number of children. The gender gap is also evident in the context of fintech adoption. [117] analysed the survey data from 28 countries and found an 8% gender gap in fintech adoption, which was higher than the 7% gender inequality in bank account ownership, where the proportion of male fintech users (29%) is significantly higher than the proportion of female fintech users (21%) in the countries due to the gender differences in users' willingness, suitability, and affordability of fintech products. Similarly, [118] pointed out that although female users in online and digital transactions in 123 developing countries from East Asia Pacific, South Asia and Sub-Saharan Africa regions lagged male users, the usage has increased owing to digital transfer of salaries and remittances. On the other hand, [119] found that fintech development reduces financial inclusion gender gap in 27 SSA countries; therefore, suggesting that the role of fintech can be strengthened through complementary targeted policy initiatives. The country-wise evidence similarly revealed the presence of gender inequality. For example, [120] conducted a study to determine the drivers and barriers of financial inclusion

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through fintech among the women in Tajikistan, and found significant differences between the levels of understanding, trust, and use of fintech between genders. In another related study, [121] carried out a cross-sectional survey in Pakistan to explore the socio-demographic characteristics of mobile banking users and observed that female users only comprised nearly one-tenth of the total users. A study conducted by [122] examining the role of social and facilitating influences and sociodemographic variables in fintech user intention in Bangladesh also provide evidence of a gender gap in fintech user intention, attributable to image, compatibility, and fintech use experience. Other related studies that investigated the contributing factors of fintech and gender gap are [123], who investigated the factors of variation in access to digital finance between men and women in 81 low- and middle-income countries, and [124], who examined the impact of fintech firms and policy shocks in reducing the gender gap in U.S. bank loans. [123] found that the main contributors to the gender gap were employment, education, and income level, while [124] showed that both fintech adoption and policy implication has significantly improved the loan approval rates of female borrowers and consequently alleviate the gender discrimination in bank loans.

In another stream of literature, researchers have studied the effect of fintech on women's entrepreneurship as the latter is usually associated with women's empowerment [125-126]. [115] reported that fintech fosters female entrepreneurs and women's empowerment in China by lowering financing threshold and operation cost and subsequently improves women's wages and narrows the gender wage gap. A similar finding was also found in Burkina Faso and Cameroon, where [127] investigated the function of fintech (mobile money) in job creation, poverty, and inequality among women. Their study revealed that mobile money has a significant positive effect on the likelihood of the women's self-employment and entrepreneurship in the two countries. Similarly, [84] demonstrated that fintech (mobile money), when complemented by banks, is more effective in encouraging marginalized populations, particularly the poor women with bank accounts, to save for microenterprises. Furthermore, [110], using data from 113 countries, showed that in countries with low gender discrimination, fintech (digital payments and online accounts) promotes women's participation in economic and entrepreneurial activities, thereby improving their living standards and financial empowerment. [116] also reported similar findings in China, where digital finance enhances women's bargaining power within households and encourages their participation in economic activities through entrepreneurial behaviour. In Indonesia, [128] surveyed a sample of 270 Indonesian women entrepreneurs and reported that fintech adoption and financial literacy are key performance drivers of women-owned small businesses. This is corroborated by the research of [129], who found a similar finding, pointing out that fintech has greater effect size than financial literacy in supporting women entrepreneurs' business sustainability in Indonesia. However, there is another group of scholars who found evidence that fintech per se is not the only determinant of women's entrepreneurship. For instance, [130] found that fintech (e-money services) in eight West African Economic and Monetary Union (WAEMU) countries have a positive influence on financial inclusion, but the effect is moderated by the presence of women entrepreneurs. Similarly, [131] examined the effect of gender, ethnicity, and geography on financing choices by comparing the data of initial coin offerings (ICOs), venture capital-backed blockchain ventures, and real estate investment trusts. The result shows evidence of enhanced financing conditions through fund raising via ICOs for ethnic minorities, sub-urban, and young entrepreneur, in addition to female entrepreneurs. This may perhaps be explained by reasons such as gender bias in financing where investors are more likely to finance entrepreneurs who are of the same gender as them, as observed by [132] in Brazil, Chile, and Mexico, when examining gender-related behavioural differences among investors and equity crowdfunding investment. [196] also found evidence that female lenders have lower confidence in cash collateral and financial literacy;

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however, their confidence increases when lending to female borrowers in pro-social P2P platforms.

The association between a specific type of fintech product, financial literacy and inclusion of women has also attracted some intention. For examples, while [133] investigated how mobile money drives financial inclusion of Nairobi women and found a significant increase in the proportion of young, less educated, and low-income women using financial services in Kenya; [134-135] examined the adoption of cryptocurrencies and crypto tokens among Spanish women and Thai investors, respectively. [134] identified several barriers in cryptocurrency adoption among women in Spain including security concern, risk aversion, and a lack of investment experience, knowledge and understanding about cryptocurrencies, whereas [135] found that gender, savings, social media, type of financial instrument ownership, education and income levels are significant antecedents of crypto token adoption in Thailand. In Ghana, [136] conducted a case study examining how a digital technology, namely, an interactive voice response system, could enable financial inclusion among low-income Ghanaian women. The IVR systems accounts for structural disadvantages that female respondents may face in using the technology and therefore can be programmed to deliver a set of customized messages tailored to their specific needs. The sample consists of 28,028 low-income female clients of a savings and loans microfinance organization in Ghana. The findings highlight the importance of user feedback, technological affordances, and environmental factors in information system design to reduce barriers to technology adoption and use. The study adds to the digital financial inclusion literature by highlighting the importance of the role of social context in the design and implementation of information system. It also presents a relational, process-oriented framework for understanding inclusion and emphasizes the significance of adopting a holistic view to understand the dynamics of inclusion and exclusion in designing, adopting, and using technology. In another research, [137] studied the opportunities and barriers of fintech services in the South Asian Association for Regional Cooperation (SAARC) and the Association of Southeast Asian Nations (ASEAN) regions. The study observed that the adoption rates of the elderly and females are relatively lower, and the demand for different types of fintech services differs across areas. From a different perspective, [138-139] paid special attention to Islamic fintech products and financial inclusion. [138] conducted a comparative analysis of the performance of fintech and financial inclusion between ten countries that adopted Islamic finance system (Bahrain, Bangladesh, Indonesia, Iran, Jordan, Kuwait, Malaysia, Pakistan, Saudi Arabia, and Turkey) and another ten countries that adopted conventional finance system (France, Germany, Ghana, India, Nigeria, Philippines, Russian, Senegal, Spain, and Taiwan). The results show that the number of fintech users is higher in countries with conventional finance, while financial inclusion and women empowerment are higher in countries with Islamic finance. Alternatively, [139] found a significant difference in the influencing factors of ZakaTech adoption between men and women in Saudi Arabia and Egypt, particularly in social influence, social isolation, and self-efficacy. This implies the need of devising strategies to improve and bridge the digital gender divide in the use of ZakaTech among Muslims. On the other hand, several researchers have examined the nexus between financial literacy and financial inclusion of women. Among them are [140], who analysed the data from 144 countries and observed a positive association between digital financial literacy and financial inclusion (engagement in formal banking channel) among women entrepreneurs, and [141], who found that digital financial literacy positively influences financial decision-making and Indian women's intention towards investment. In Indonesia, [128-129] found that financial literacy supports Indonesian women entrepreneurship, while [142-143] provide evidence of indirect effect of financial literacy on fintech adoption intention among Indonesian women. However, a few studies reported that women's financial literacy is relatively lower than men. For instance, [144] examined the role of financial knowledge in the use of fintech services in Vietnam

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and found that women possess lower levels of actual and perceived financial knowledge and have less propensity to use fintech services than men. Similarly in Saudi Arabia and Nigeria, [145] observed that Saudi women only possess a satisfactory level of basic financial literacy, while [146] revealed that Nigerian women have less interest in complex financial services due to barriers in trust and digital literacy.

In general, literature has on the one hand demonstrated that the digital finance reduces gender inequality [109, 116], but on the other hand, pointed out the persistence of gender inequality in financial access and financial inclusion, highlighting the difficulties faced by women due to barriers in socio-economic [112], education and financial literacy [144-146], employment and income level [114, 123, 135], and trust in fintech [120]. More specifically, [117] explained that the existence of fintech gender gap may be attributable to the attitudes of females towards new fintech subject to its suitability and affordability, whereas [123] viewed the digital financial inclusion disparity between genders as the result of the gaps in the levels of employment, education, and income, and [146] argued that women may be hindered from the complex financial services because of trust and digital literacy barriers. [134] also offered reasons for the gender gap in cryptocurrency investments, such as females generally have less investment experience and knowledge in cryptocurrencies and are more risk averse than males. Alternatively, literature also corroborates the view that fintech enables women entrepreneurship thereby improves their financial inclusion and reduces gender inequality [116, 127, 129]. Besides, [111] found that high gender inequality is associated with low female financial inclusion and hence highlighted the importance of improving the level of education, employment, and political participation of women to improve their financial inclusion. [119] also documented that fintech development narrows the gender gap in financial inclusion and should be supported by specifically tailored complementary policy and programs aimed at reducing gender inequality. In sum, research have also shown that women in marginalized communities or rural areas often face greater barriers in fintech adoption than urban women due to limited internet infrastructure, low ownership of digital devices, and socio-cultural norms that restrict women's role within the family and limit their financial autonomy [245-247]. Additionally, women in developing countries frequently have limited exposure to formal education and financial knowledge, which contribute to their lower levels of financial literacy and greater barriers to fintech adoption [144]. In other words, rural women in developing countries with greater gender discrimination are likely to face much greater challenges in adopting fintech and financial inclusion due to lower digital financial literacy and socio-economic status [111-114, 144-146].Considering that fintech alone may not be sufficient to close the gender gap and improve women's financial inclusion, a broader set of complementary interventions, such as targeted initiatives and strategies promoting gender equity in education and employment, as well as the development of digital infrastructure is necessary to reduce gender inequality in digital financial inclusion and to enable women's empowerment [107, 133].

4.5 Fintech, financial inclusion and rural communities

The financial inclusion of the rural communities through fintech is also one of the central topics among researchers (see Appendix E), as earlier research has shown evidence of the presence of heterogeneity in incomes and financial inclusion between urban and rural communities [147-148]. For instance, [149] pointed out that poor, non-white, and indigenous rural communities have the lowest rates of fintech in the U.S. due to digital redlining, while poor white rural communities report relatively higher fintech rates, as they are better equipped with high-speed internet access and comparatively have more accumulated wealth. Although several studies have raised doubts about the effectiveness of digital technologies

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in improving financial inclusion and incomes of rural communities, the overall literature still provides evidence that supports the role of fintech in promoting financial inclusion and reducing the rural-urban gaps. More specifically, [150] reviewed how digital financial inclusion has changed rural digital financial system in China using the models of two China Internet giants (JD and Alibaba). They argued that although digital financial inclusion could benefit rural populations and companies, it is still uncertain if this will bring improvement to the rural communities. [151] also conducted a study to find out how financial services in rural markets in developing countries are reshaped by digital technologies. Their finding pointed out not only the imperfections of current digital technologies but also highlighted that these digital technologies are likely to improve the efficiency of the rural financial system over time. Similarly, [147]'s finding from a panel data of 202 cities in China also indicated that digital economy improved income levels of the people. However, the positive effect is greater in urban areas and therefore worsen the income gap between urban and rural areas. In contrast, [152] used panel data to examine the role of fintech in Chinese rural prosperity and documented empirical evidence that fintech significantly improved farmers' prosperity, while [153] showed that digital inclusive finance is essential in promoting rural revitalization in China. Also, [154] found that fintech improved wage and property incomes of rural households when examining the relationship between fintech adoption and rural economic development in China. Furthermore, [155] found digital finance inclusion could facilitate poor and rural household consumption in China, and [156] observed that mobile payment has a significant positive effect on the consumption of Chinese rural households, mainly attributable to the benefits and lower costs of mobile payment, which promote rural commodity transactions and financial inclusion. Other significant effects of fintech and digital financial development on rural communities are facilitating the decision making of Chinese rural to urban migrants who are settling down for the long-term [157], promoting individuals' charitable giving behaviour through increasing residents' internet usage, income, and windfall income, particularly in rural areas and eastern China [158], lowering the possibility of food and health poverty, especially among the rural populations in Kenya [159], and reducing Ghanian rural households' diversification in crop production while increasing non-farm businesses and livestock production as additional livelihood options [160].

The underlying relationship between fintech, financial inclusion, and rural entrepreneurship, particularly in China has also attracted much attention from researchers. For example, [161] used the data of rural entrepreneurs' households and found that online transactions have substantial direct and indirect effects on digital financial inclusion of Chinese farmers, especially new entrepreneurial farmers who are highly educated and skilled. [162] also showed that digital finance development facilitates urban-rural integration in China through entrepreneurship, especially in the eastern regions and areas with low human capital. Additionally, [163] found that through entrepreneurship, the urban-rural income gap could be significantly reduced by the coverage but not the depth of digital financial inclusion. This is of particular significance in less educated and underdeveloped regions. [164], when investigating the function of digital financial capability in online entrepreneurship in China, also found that digital financial capability improved rural household entrepreneurship and digital business performance of enterprises, particularly in retailing and food and beverage sectors. Similarly, [165] examined the nexus between digital inclusive finance and entrepreneurial activities in Chinese rural areas and observed that digital inclusive finance promotes entrepreneurship by facilitating credit and financial information. In another related study, [166] used the panel data from the Digital Financial Inclusion Index (DFI Index), and China Household Finance Survey (CHFS) to investigate the effect of digital finance on rural entrepreneurs' rate of return and found a positive relationship owing to the facilitated external financing and reduced cost of financial transactions. More specifically, [57] found that the adoption of mobile money increases the likelihood of engaging in entrepreneurship

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by 24.4% among rural women and youth by in Kenya, Tanzania and Uganda. Also, through interviews and qualitative analysis, [250] summarizes several benefits of using mobile money among Kenyan micro-entrepreneurs, including timely updates and convenience for savings, payments and securing loans. The finding is in line with [252], which reports that mobile money has improved and transformed the livelihoods of rural communities in Cambodia implying that mobile money services exhibit strong potential to enhance micro-entrepreneurship in the country. Additionally, [241] conducted a bibliometric analysis on the role of fintech, such as mobile banking and microfinance, in enhancing financial inclusion of underserved communities in emerging markets. The review indicates that fintech not only facilitates financial inclusion but also enables online financial transactions and improves access to credits, thereby fostering economic resilience and growth in rural areas. This is corroborated by another systematic review, which showed that P2P lending enhances financial inclusion and supports the credit access for small and medium enterprises (SMEs) in developing countries through technologies like artificial intelligence and blockchain [249]. Other relevant and interesting studies include [167] who revealed that digital finance non-linearly enabled rural revitalization in China through improved technological innovation, entrepreneurial activity, marketability, and accumulation of human capital, and [168] who provides empirical evidence of digital finance development affecting intergenerational income mobility in China, in which digital finance encourages entrepreneurship and thus leads to lower intergenerational income correlation.

Considerable research has also been conducted to examine the antecedents of digital financial inclusion and fintech adoption in rural communities. From their literature review, both [241] and [249] have unanimously highlight the importance of internet connectivity and other digital infrastructures in enabling fintech for rural development. Without adequate infrastructure, rural communities may continue to face significant barriers despite the potential benefits of fintech. The importance of fintech infrastructure and access is also emphasized in several studies focusing on the context of China. Among others, [171] demonstrates that digital finance and infrastructure support rural revitalization and development in China, while [172] reveals that the extensive construction of broadband infrastructure in rural China significantly contributes to the coverage but not the usage of digital finance in regions with greater human and social capital and more bank branches. [169] conducted an in-depth analysis of Indian government policies and fintech services and suggested financial inclusion in India can be achieved not only through the development of fintech infrastructures but also through government incentives to encourage its citizens, particularly those who reside in Indian suburban and rural areas to adopt the fintech services. This echoed the study of [170], which highlighted that although the Indian government has supported and provided the rural masses an easy access to financial products, the corresponding policies and governance that match the speed of financial innovation are also required ensuring secure and transparent development of financial inclusion. Also, [173] used a spatial measurement model to evaluate the effect of digital inclusive finance on financial inclusion in China and observed that advanced digital technologies widen and ease accessibility to financial services and thus expand financial inclusion to rural areas. Likewise, [174] also found that young, high-income, and better educated consumers in rural China are more willing to use mobile payment, conditional on the availability of more smartphones and financial services, a high perception of mobile payment adoption by their neighbours, and social or distance constraints when visiting banks. Similarly in other countries, [175] observed that mobile money users in rural Uganda are more willing to use financial services, such as saving, borrowing, and receiving remittances, demonstrating the necessity of improved access to pro-poor financial products in accelerating financial inclusion of rural communities, whereas [176] examined the function of P2P in promoting financial inclusion in the U.S. and found that P2P loan requests increased if there is a decrease in the number of urban pawnshops and rural bank branches.

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Another stream of literature has focused on the adoption process of fintech among rural communities. Despite having access, rural populations may still be reluctant to adopt fintech due to other reasons. Therefore, it is vital to gain better insights into the issue from functional, behavioural, and psychological perspectives on fintech adoption and financial inclusion. [251] explores the challenges associated with fintech adoption among rural populations and highlights several issues, including limited access to technology, low financial awareness and lack of trust. Using Nigeria as a case study, [244] conducted a content analysis to investigate the persistent financial inclusion gap, despite various efforts by the governments and financial institutions. The result identifies several contributing factors, including poor infrastructure, intermittent power supply, high costs, data privacy breaches, low financial literacy, and information asymmetry. [177] conducted a literature review and found that security, trust, convenience, social influence, perceived ease of use, perceived risk, and perceived usefulness are influential factors affecting farmers' fintech adoption intention, and these relationships are moderated by demographic factors such as gender, age, and experience. In the case of China, [178] developed a conceptual framework to examine the cause and effect of using digital finance platforms by rural Chinese in managing their personal finance, and found that the adoption of digital finance is determined by performance expectancy, effort expectancy, facilitating conditions, and perceived risk, while [179] investigated the antecedents of fintech adoption among rural Chinese and demonstrated that perceived usefulness, perceived ease of use, innovation awareness, and financial awareness are significant determinants of fintech application adoption intention. [180] also found that active online social interaction, especially via contextual activity, has a positive effect on the digital finance involvement of Chinese rural households. On the other hand, [181] found that subjective norms, perceived security, and technology self-efficacy are among the significant factors affecting rural Omani millennials decision to use of mobile wallet, whereas [182] showed that compatibility, complexity, perceived agent trust, perceived risk, and facilitating conditions are significant factors of fintech adoption in rural and remote areas in Tanzania. Along the same lines, financial literacy is also found to be an influential factor of rural financial inclusion. [14] conducted a review on financial inclusion in rural India and concluded that financial illiteracy, among other barriers (e.g., high cost, geographical access, and inappropriate banking products) must be overcome to promote financial inclusion in the country. The finding is consistent with the results that showed financial literacy is a dominating factor in promoting access to fintech services and accelerating financial access and inclusion among the rural population in Bangladesh [183-184]. Likewise, [185] examined the impact of digital financial literacy on income inequality in rural areas of China and found that digital financial literacy reduced income inequality, and [186] also found that knowledge of inclusive finance improved Chinese farmers' perceived benefits and financing intention of ordering finance (a special loan service for farmers) in China.

To summarize, the literature showed that the urban-rural gap characterized by accessibility is progressively being filled following promising development of fintech and digital infrastructure in rural areas, which in turn contribute to improved financial inclusion and economic empowerment of rural communities [147, 153, 171]. While poor rural populations in some regions may still experience digital redlining and urban-rural divide, the marginalization and income inequality can nevertheless be mitigated through rigorous policy intervention, enhanced fintech penetration and coverage, and improved levels of entrepreneurship [149, 154, 162-163, 167]. Specifically, entrepreneurial returns and activities in rural areas can be stimulated through digital finance inclusion and individuals' ability of using digital finance, particularly via mechanisms such as enhanced accessibility to financial services, and reduced financial information barriers, credit constraints, and transaction costs [164-166]. In view that the poor and rural populations are frequently excluded from fintech because of their limited knowledge and capability

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in understanding and using fintech [180], digital financial literacy and education were also highlighted in literature as a contributing factor that can reduce urban-rural digital divide and improve the digital financial inclusion and development of rural areas [156, 182, 184-185]. Other contributing factors to digital financial inclusion including functional, psychological, and behavioural factors such as perceived usefulness, perceived ease of use, social influence, facilitating conditions, and trust [177-178, 182].

4.6 Fintech, financial inclusion and other minority vulnerable groups

Research concerning the fintech adoption and financial inclusion of children and young people, people with disabilities, the elderly, migrants, and other minorities was also documented in the literature (see Appendix F). For example, a group of scholars has provided evidence of the links between fintech, financial inclusion, and children or young people across many countries. [187] reported a widespread adoption of digital financial services among semi-urban Ghanaian university students and thus suggested that digital financial services can be a pathway for financial inclusion as they facilitate job creation and small enterprise development, particularly in Ghana's semi-urban communities. In Vietnam, [68] surveyed a sample of 1288 students to understand the effect of fintech and financial literacy on the financial inclusion of young Vietnamese. The result revealed that fintech and financial literacy accelerated financial inclusion development in the country, and there is relatively easy access to financial services in Vietnam despite the urban-rural gap in access rate between the two areas. In another related study, [188] surveyed a sample of 689 Gen Z consumers and found that the reuse intention of digital financial services depends on their understanding of technology and digital finance, attitude toward consumer protection law enforcement, and perceived risk of using digital financial services. A few studies also examined the factors influencing the fintech adoption of young people. These include [189], who observed from a sample of 397 Indian college students that fintech (AI) adoption was positively influenced by perceived usefulness, satisfaction and ease of use, while negatively affected by concerns about privacy and security; [256] found, based on a sample of 158 Gen Z and 104 Gen Y in India, that the behavioural intention to adopt digital payment fintech services depends on effort expectancy, performance expectancy and customer satisfaction; [190], who found a high level of perceived usefulness, ease of use, and security would increase the e-wallet usage intention among Malaysian youth; and [191], who analysed the data collected from 166 students across fifteen higher education institutions in Brazil, found that performance expectancy, effort expectancy, and security had significant positive effects on the Brazilian students' intention to use fintech services. The relationship between digital financial literacy and financial technologies usage among children and young people was also investigated. [192] carried out a qualitative study on 22 focus groups of children aged between 8-10 years old in Mexico and found that children are aware of the need for digital financial education to understand critical concepts of digital finance, digital finance security, online financial operations, and the use of mobile applications. [193] also examined the relationship between digital financial literacy and the adoption of cashless payment systems among university students in Jordan, and demonstrated that digital financial experience, awareness and skills are the main antecedents of cashless payment adoption. Researchers have also focused on the interest of young people in P2P lending. For example, [194] found a direct positive effect from the attitude, and indirect effects of perceived ease of use, knowledge, and trust on the intention to invest in P2P lending by Indonesian Muslim youths with high levels of income and education. Additionally, [195] used a mock blockchain-based P2P site to examine 909 lending decisions by 303 finance students in the U.S. and found that students with financial trauma experience are more willing to lend more to highly trusted applicants, particularly among the male students who are more inclined to fund more credible female loan applicants.

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A recent strand of literature has paid attention to the impact of fintech and financial inclusion on people with disabilities and the elderly. For example, in India, [197] interviewed a sample of 30 respondents with visual impairment disabilities and revealed that they encountered difficulties in using digital banking because of social and technical challenges. This is in line with the findings of [198], which highlights that lack of financial education, digital infrastructure, diverse financial products, and financial law with accessibility provisions, are among the barriers that hindered the people with disabilities from using the digital financial services in China, and [199], who found that people with disabilities in Ghana are interested to use mobile money if they face constraints in accessing and using conventional financial services. The above research was echoed by [200-201], who further examined the determinants of fintech adoption among people with disabilities in India. More specifically, [200] surveyed a sample of 205 Indian respondents with disabilities and found that fintech adoption indirectly improve their financial wellbeing through enhanced financial access, knowledge, and behaviour, while [201] used the data of 323 Indian respondents with disabilities and showed that all factors of the cognitive, affective and normative model, except the perceived complexity and technological anxiety, have a significant positive effect on the digital insurance adoption intention of people with disabilities. In another stream of research, [202] investigated the effect of digital financial inclusion on Chinese elderly consumption expenditure and revealed that digital financial inclusion stimulates the elderly's consumption attributed to better financial service accessibility. Relatedly, [203] analysed the characteristics of fintech and media users in South Korean and observed that the elderly people (age 50-64) who are staying together with younger generations tend to use more fintech, and [204] surveyed a sample of 425 Chinese elderly respondents and found that the mobile banking engagement by the elderly were directly affected by perceived ease of use and perceived usefulness of mobile banking platforms and indirectly influenced peer engagement and word-of-mouth through perceived usefulness. In another research, [205] examined the data of China Health and Retirement Longitudinal Study (CHRLS) and observed a significant positive relationship between digital financial inclusion and subjective well-being (depression and self-reported health status) of the elderly in China, but in contrast, [67] found a negative correlation between digital finance and life satisfaction of Chinese elderly people (over 60 years old) when examining the China Health and Pension Tracking Survey (CHPTS) data. They explained this as the result of ensuing disappointment and frustration from struggling with digital financial services, and a lack of emotional comfort and companionship from their children [206].

Research concerning minorities such as migrants and ethnic minorities was not neglected in the literature as scholars have also examined how fintech and financial inclusion affect these minorities. For examples, [207] studied the subject from the racialized dimension of fintech in improving the financial inclusion of refugees in Kenya and concluded that the refugees in the country are still largely financially excluded. [208] also found a racial gap in financial services and wealth accumulation, where the Black, Latinx, and other minorities have to pay more for the services compared to the White. In Bangladesh, [209] observed that training designed to promote mobile banking and technology adoption in family networks by poor and illiterate Bangladeshi families showed a significant increase in mobile banking usage by migrants who remitted money back to their extended families. In another related study, [210] interviewed 40 Roma from Romania who are homeless in Denmark to understand how they use digital banking cards. The finding suggested that in order to support the digital financial and economic inclusion of marginalized communities, digital financial initiatives must be implemented by considering all the five aspects of access, that is, the availability, accessibility, accommodation, affordability, and acceptability of access. Furthermore, [211] conducted interviews with 24 professionals from institutional, state and private sectors, and 28 remittance recipients in Ghana, and found that

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fintech services such as digital remittances and mobile money have improved the interviewees' financial decision making and wellbeing, owing to better promotion and advice services, as well as expedited and simplified financial processing with minimal paperwork. Relevant research was also conducted in China. For instance, [212] studied the effect of digital financial inclusion on income levels of herdsmen in Chinese ethnic minority provinces and found that the degree of digitalization, extent of coverage and depth of usage increase the income of herdsmen. Also, [213] used the data from the China Migrants Dynamic Survey (CMDS) to study the effect of digital finance on migrants' entrepreneurship and found that digital finance accelerates low-income survival-oriented entrepreneurship of migrants to maintain their livelihoods. In a similar study, [214] also used data from the CMDS, along with the DFI index, and reported a positive association between the local digital financial development and entrepreneurial propensity and quality of migrants. Specifically, the result showed the digital finance digitization level, coverage breadth, and depth of use facilitated Chinese migrants' entrepreneurship, particularly for less educated rural migrants without local hukou, party membership, and cross-provincial mobility experience. In another study, [215] analysed data from the China Family Panel Studies (CFPS), and the Digital Inclusive Finance (DIF) Index to examine how green digital finance development affects the Chinese residents' migration decisions considering that green digital finance improves life quality and promotes migration. The result revealed that migration decisions were positively influenced by green digital finance and thus implies that migrants would be more interested and choose to reside in areas with more and better green digital finance services.

Overall, the literature showed that fintech facilitates digital financial inclusion and wellbeing of the young people, the elderly, and minorities (e.g., people with disabilities and migrants) by reducing barriers to financial services that meet personalized financial needs [68, 200, 202, 209]. The younger generations face challenges in adopting fintech due to security and digital financial literacy issues [187, 192, 194], while the elderly often refused to adopt fintech because of insufficient digital literacy, lack of trust, and complicated fintech products [64, 203-204]. On the other hands, people with disabilities often face both social and technical barriers in fintech adoption, such as inconvenient access and unsatisfactory assistive technologies [197-198], whereas the systemic inequalities and socio-economic marginalization, such as formal identification, language barriers, and discriminatory practices in financial systems often hindered ethnic minorities and migrants from fintech adoption [214]. Intersectionally, [52] and [53] observed that poor, less educated young people in Kenya and Western African Economic and Monetary Union region, respectively, are financially excluded and gain little benefit from fintech. However, [63] pointed out that age and education level are moderators of the intention and usage of mobile payments in Ghana. [47, 48] and [50] also documented that rural and less educated elderly, and lower social class communities in India and the MENA region, respectively, are less likely to adopt and benefit from fintech. However, [64] highlighted that the adoption of digital finance services among the rural elderly, such as in China, can be enhanced through higher financial literacy. The literature also documented that younger generations are more likely to adopt a fintech product or service if they have high perceived performance expectancy, effort expectancy, reliability, security and trust on the digital financial platform [187-188, 191, 194]. The level of digital financial literacy is also found to be one of the main determinants of their intention and decision to use fintech, particularly among the young people in rural areas [68, 192]. Similarly, the adoption of fintech by elderly people is also influenced by the perceived usefulness, ease of use, security, trust, and digital use experiences [203, 216]. However, the literature also revealed that digital financial inclusion on the one hand improved the elderly's overall self-reported health and depression conditions [205], but on the other hand reduced their emotional comfort and life satisfaction [67]. Previous studies have also reported that fintech such as digital banking and mobile money has improved the accessibility of people with

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disabilities to digital financial services and reduced the inequalities by reducing disability-related barriers in commuting, documenting and communicating [199-200], despite some of them may still encounter denied access and be hindered by the complexity of the financial technology [197, 201]. Therefore, efforts are needed to ensure that the readily and easily accessible digital financial services are being made available to foster a more digitally enriched quality of life and welfare of people with disabilities [198]. Besides, literature also shows that fintech enabled survival-oriented entrepreneurship and easier access to entrepreneurial finance among disadvantaged and low-income migrants [213-214], and fintech adoption among migrants can be accelerated through well-designed interventions [209]. Both [210, 212] also demonstrated the essential of accessibility to digital financial services for the minorities and marginalized communities in progress towards the betterment of their livelihoods and social inclusion.

5. Future research directions

The findings and conclusions of extant studies in fintech, financial inclusion, and vulnerable groups were not without limitations, and researchers have accordingly proposed suggestions for future studies to extend the investigation in these fields. Data type, measurements, collection methods, and generalizability of findings are the few common limitations highlighted in previous research. For example, both [185, 189] highlighted that cross-sectional data may not reflect the dynamic nature of the data and variations in the relationships over time. Therefore, it would be worthwhile for future studies to consider using panel data or longitudinal studies to obtain a more comprehensive understanding of the dynamic associations between the variables of interest. Other researchers who also encountered similar limitations and proposed the same suggestion for future research are [60, 67, 102, 129]. In addition, previous research also faced limitations in the extensiveness and accuracy of variable coverage and measurement and has proposed some suggestions for future research consideration. For instance, both [145, 186] suggested that future research may consider improved measurements of embeddedness of social enterprise and digital finance, and financial literacy, respectively, while [73] suggested including additional indicators to improve the comprehensiveness of the financial inclusion index, and [77] suggested using multi-dimensions of information, system, and service qualities instead of the unidimensional scale used in their study. [178] also highlighted that the construct of perceived risk used in their study may not be completely or sufficiently measured, and hence future research may consider other alternative measurement items. Another common limitation of previous studies is that structured survey or questionnaire-based research may not fully capture the nuanced aspects of responses such as subjective insights and non-verbal signals. Accordingly, [68, 128, 130, 179, 204], among others, have suggested that future research may consider more interactive methods such as one-on-one and in-depth interviews, mixed-methods, experiments, or focus group discussions to achieve a more holistic and depth of understanding of the research topic. Furthermore, geographic and demographic constraints have also limited the generalization of some earlier findings. For example, [181] noted that their finding was limited to rural Omani millennials and thus suggested that future research should consider widening the focus of research covering more diverse set of respondents and areas. The same suggestion was also put forward by [80, 107, 114, 123, 167], that it would be of great significance to compare and provide more diverse and universal international evidence from regions and countries with different levels of development.

Additionally, the present review has opened several new avenues and directions for future research. First, there is an urgent need to address the growing challenge of digital fraud and risk of cybersecurity breaches on digital platforms as it will adversely affect fintech adoption and digital financial inclusion, particularly among

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the vulnerable populations who are financially more vulnerable [141]. Although the emergence of fintech has improved and made financial services more efficient and accessible [28-29, 31], it has at the same time, intensified the concerns about risks of cybersecurity vulnerabilities, such as cybercrimes and online fraud in fintech [217-218]. Most literature has centred on the benefits of fintech but there is also a need to delve into the negative factors and impacts of fintech adoption amid increased cybersecurity breaches and online fraud incidents [219-220]. Cybersecurity has a major impact on fintech development [221]. Moreover, [222] showed that security risk concern is one of the main obstacles to fintech adoption as potential users may be swayed away from fintech due to worry and fear of weak security measures, hacking, and fraud, which result in possible personal data breaches, leakage of sensitive information, and financial loss. Also, [46] showed that although mobile payments could increase people's happiness, the non-cash payment system also caused over-consumption that undermined happiness. They also highlighted other adverse aspects of mobile payments, such as privacy concerns and the digital divide, which may also threaten people's welfare and thus warrant further investigation. Fraud in relation to fintech lending [74] and fintech's risk perception [53] are areas that also require further research attention. In sum, fintech service providers must ensure that the fintech platforms are configured with trustable and reliable cybersecurity and fraud detection systems, and therefore future research may analyse the impact of risks of using fintech services, such as cybersecurity vulnerabilities and frauds, on fintech adoption and financial inclusion, focusing specifically on vulnerable groups.

Second, although there are many studies on fintech and financial inclusion of women, poverty, and rural communities, research concerning the elderly and other minorities, such as people with disabilities and migrants is still limited. The present literature revealed that digital financial inclusion, on the one hand, improved the elderly's subjective well-being [205] and consumption [202], but on the other hand, reduced their life satisfaction [67]. Therefore, more in-depth investigation is necessary to better understand the impact of digital financial inclusion on the elderly people. While [203] suggested that future studies may consider including socio-environmental, psychological, regulatory and/or mediating factors involving fintech adoption of the elderly, [67] have suggested exploring potential mediating factors, such as social support, cognitive functioning, and health status. Future research may also consider the function of digital financial literacy within the framework of elderly communities as high digital financial literacy was shown to improve financial well-being, owing to the greater ability to guard against digital fraud [223-224]. Another vulnerable minority group that deserves more attention from scholars is people with disabilities. Empirical evidence has revealed that people with disabilities are still largely excluded from financial services [199]. Although [200] showed that fintech adoption can indirectly improve the financial well-being of people with disabilities through better financial access, knowledge, and behaviour, [197] found that people with visual impairment disabilities still encountered difficulty in accessing digital financial services owing to social and technological challenges. Furthermore, [198] pointed out that access to digital financial services for people with disabilities was mainly hindered by the absence of digital infrastructure, financial education, diverse financial products, and financial law with accessibility provisions. Therefore, future studies may consider focusing on specific disabilities and consider other factors such as privacy, trust, and customer satisfaction [201]. Also, future research may consider examining the role of regulatory policy and support, and digital financial literacy in improving digital financial inclusion of people with disabilities [225]. Likewise, the digital financial inclusion and welfare of other minorities should not be neglected either. While some studies reported that fintech and digital financial inclusion improved financial wellbeing and livelihoods of the ethnic minorities and migrants [211-213], other research has demonstrated that the racial gap in financial inclusion persists, in which the ethnic minorities and migrants are still largely financially excluded [207-208]. Therefore, future research may consider

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[210]'s suggestion to include all five aspects of access in digital financial initiatives to facilitate the digital financial and economic inclusion of the minorities and marginalized communities. Further in-depth investigations may also be carried out by incorporating psychological and socio-economic factors, as well as mediating or moderating effects of demographic factors and many other interesting factors such as financial literacy and financial services mistreatment [211, 226-228].

Third, future studies could consider extending the literature with more research on fintech and traditional finance. While it is undeniable that fintech plays a key role in driving financial inclusion, the function of traditional finance should not be disregarded as it is unlikely disappear entirely since not everyone is technology-savvy or has easy access to digital finance, particularly the elderly, less educated and rural communities [157]. Also, [155] showed that both digital finance and formal finance are influencing factors in the consumption among the poor and rural households. Therefore, future research may consider examining the interplay between fintech, traditional formal and informal finance as [36] pointed out that perfect financial inclusion will ultimately lead to unified integration of formal and informal financial sectors. Furthermore, [127] also suggested that future studies may investigate the integration or substitution of fintech and traditional finance. [229] demonstrated the importance of the dynamic relationship of formal and informal finance in developing financial inclusion strategies as individuals have the choice between formal and informal finance channels subject to their social networks and circumstances. More specifically, the study showed that wealthier households with close kinship ties are more inclined towards informal lending, whereas those with wider social networks, higher income and assets preferred the formal lending channel. In addition, research has also discovered significant positive bi-directional relationship between fintech and traditional finance, where traditional finance has been shown to accelerate the adoption of fintech and vice versa, while fintech has likewise facilitated traditional finance to innovate and adapt to the changing financial landscape following the technological advancements [230]. In a related study, [231] investigated how fintech credit affects bank lending, and how the relationship was affected by bank branch networks. They found that fintech complemented traditional bank lending but had a negative net effect on bank lending if the branches of traditional banks were reduced. This implies, in particular, the importance of physical bank branches for financial inclusion in areas with inadequate digital infrastructure. They also suggested that future research may consider examining reasons for the varying competitive and complementary effects, and how banks can leverage fintech for greater financial inclusion. Therefore, further in-depth studies on the complementary or substitute relationship between fintech and traditional formal and informal finance in financial inclusion would supplement the literature with better insights into the complex relationship.

Lastly, the role of regulators in digital financial inclusion of vulnerable groups is also worth further exploration. Researchers have suggested that future research may consider investigating the influence of institutional environment and the function of regulators within the ecosystem of fintech development by including variables such as regulatory framework and trust in government policies in the context of fintech adoption [54, 111, 173, 179, 181]. Another researcher, [119] also suggested that future studies may examine the moderating and complementary effects of policy initiatives. [123, 222] also pointed out that in countries with growing fintech and transitional regulatory frameworks, the concerns about weak cybersecurity frameworks, fraud, and monetary loss could potentially adversely impact digital financial inclusion. Furthermore, [232] have highlighted that rapid advancement of the fintech industry has surpassed the development of robust regulatory frameworks, and hence the likelihood of misbehaviour and fraudulent activities in fintech would be high in the absence of an effective regulatory frameworks and supervision mechanisms. Accordingly, future research may consider examining the function of regulatory and

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supervisory technologies in facilitating fintech's compliance with changing regulatory obligations and regulators' surveillance and enforcement capabilities, which in turn accelerate digital financial inclusion [233]. Therefore, considering that the rapid growth and change in fintech have transformed the global financial landscape, and thus presenting opportunities as well as challenges for the fintech industry, regulators, and investors alike, it would be valuable for future research to complement the growing literature on fintech, financial inclusion, and vulnerable groups with meaningful insights from the perspective of the role of regulators, regulatory frameworks in shaping the fintech industry and promoting digital financial inclusion among the vulnerable groups. The summary of limitations and suggestions of previous studies, and the research gaps and future research directions is summarised in Table 3.

Table 3 The summary of limitations and suggestions of previous studies, and future research directions

Panel A The summary of limitations and suggestions of previous studies		
Limitations	Suggestions	References
Cross-sectional data cannot reflect the dynamic changes in the relationships.	Consider panel data or longitudinal studies.	[67, 102, 129, 185, 189]
The extensiveness and accuracy of variable coverage and measurement.	Consider improved measurements, additional indicators, alternative measurement items, and/or multi-dimensional data.	[73, 77, 145, 178]
Structured survey or questionnaire-based research cannot fully capture the nuanced aspects of responses.	Consider one-on-one and in-depth interviews, mixed-methods, experiments, and/or focus group discussions.	[68, 128, 130, 179, 204]
Geographic and demographic constraints.	Consider more diverse set of respondents, regions and countries.	[78, 80, 107, 114, 123, 181, 185]

Panel B The research gaps and future research directions		
Research gap	Future research direction	References
Growing challenge of digital fraud and risk of cybersecurity breaches on digital platforms, and intensified concerns about risks of cybersecurity vulnerabilities	Future research may examine the impact of risks of using fintech services (e.g., cybersecurity vulnerabilities, frauds, personal data breaches, leakage of sensitive information and monetary loss) on fintech adoption and financial inclusion of vulnerable groups.	[46, 53, 74, 217-218, 222]
Lack of research on the fintech and financial inclusion of the elderly and other minorities, such as people with disabilities and migrants.	Future research may conduct more studies to further understand the impact of digital financial inclusion on the minorities and marginalized communities by examining direct, mediating and/or moderating effects of regulatory, psychological, functional and socio-economic factors.	[67, 200-201, 203, 210, 212, 223-228]
Limited scholarly attention on the interaction between fintech and traditional finance in relation to vulnerable groups.	Future research may consider examining the relationship between fintech, traditional formal and informal finance such as the interaction and bi-directional relationships, and the integration or substitution of fintech and traditional finance.	[36, 127, 155, 157, 229-231]
The role and function of regulatory framework and regulators in digital financial inclusion of vulnerable groups remains room for further investigations amid the rapid	Future research may investigate the effectiveness of regulatory framework and efficiency of regulators within the ecosystem of fintech development by considering factors such as regulatory	[111, 119, 123, 173, 179, 181, 222, 232]

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growth and change in fintech, which created opportunities as well as challenges.	framework, role of regulators, regulatory and supervisory mechanisms and technologies, and trust in government policies.	

6. Conclusion

While existing research has conducted topic- or geographic-specific reviews and bibliometric analysis on fintech and financial inclusion, the review on the nexus between fintech, financial inclusion, and vulnerable groups is still insufficiently explored. Therefore, this paper systematically reviews 185 peer-reviewed journal articles in relation to fintech, financial inclusion, and vulnerable groups. The reviewed articles were published in 136 indexed journals between 2017 and 2024, involving 160 authors and affiliations from 64 countries. The result showed that research concerning fintech, financial inclusion, women, poverty, and rural communities has received the most attention, whereas topics concerning people with disabilities and elderly people have garnered the least focus. Also, nearly 75.14% of the research involved quantitative data analysis, while the remaining used other approaches such as qualitative, mixed-method, experimental, descriptive, and literature review. There is also an increasing trend of publications and citations since 2017, particularly between 2021-2024, in which the number of publications and citations has reported a marked increase of more than fivefold and eightfold within the three years. In addition, the limitations and suggestions of previous studies were summarized, and four main research directions that have great potential and may interest future studies were identified from the review. These include addressing, first, the growing challenge of digital fraud, risk of cybersecurity breaches, and intensified concerns about risks of cybersecurity vulnerabilities to understand the impact of risks related to fintech use on fintech adoption and financial inclusion of vulnerable groups. The elderly, less educated, low-income and rural populations are most disproportionately affected due to limited access to secure technologies and digital financial literacy. Therefore, there is an urgent need for studies that specifically focusing on cybersecurity vulnerabilities and digital literacy among these groups. Specifically tailored cybersecurity policies, and targeted awareness and education programs that consider the unique vulnerabilities of these marginalized groups will reduce exposure to digital threats and enhance trust in digital financial services. Second, the need for more research on the fintech and financial inclusion among the minorities, such as the elderly, people with disabilities, and migrants to understand the impact of digital financial inclusion on these marginalized communities. Currently, there is a lack of in-depth understanding how these fintech and digital financial inclusion affect the minorities, who typically encounter greater barriers, particularly in terms of accessibility and inequality. Therefore, context-specific research on inclusive design of fintech, regulatory frameworks, and social and financial inclusion, focusing specifically on the minorities is highly encouraged. Third, the relationship between fintech and both traditional formal and informal finance in the context of vulnerable groups, with the aim of understanding the interaction and bi-directional relationships, and the integration or substitution of fintech and traditional finance. Further research in this area is particularly important to underserved populations and regions. Fintech offers more convenient and faster financial services, while traditional finance is more reliable and trustworthy. Therefore, an in-depth understanding on the complementary or substitutive relationship or bi-directional relationships between fintech and traditional finance, and how they can be blended into a hybrid financial model that supported vulnerable groups would be particularly valuable. Fourth, the role and function of the regulatory framework and regulators in assessing the effectiveness of regulatory framework and efficiency of the regulatory and supervisory mechanisms and technologies in digital financial inclusion of vulnerable groups.

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The rapid growth of fintech has outpaced regulatory and cybersecurity frameworks, exposing vulnerable groups to greater digital threats. Therefore, research on regulatory framework and tools to enhance regulatory efficiency and monitoring to enable the government to keep pace with technological advancements and better safeguard and serve the marginalized populations would be of great value. This study highlights the important role of governments in the development of fintech and financial inclusion for vulnerable groups, particularly in developing inclusive fintech policies and digital infrastructure, promoting digital financial education, and fostering public-private collaborations to create robust privacy and cybersecurity protection. Meanwhile, fintech companies also play a vital role in providing inclusive digital platforms that are easily accessible, less complex, and tailored with personalized financial assistance and advice, to meet the specific needs of the underserved and marginalized populations. Financial institutions may consider partnering with fintech providers to develop hybrid fintech models that blend fintech with traditional finance, offering more accessible, flexible, and trustworthy financial products and services to the vulnerable groups. Lastly, while this study contributes to the body of literature in this regard, it is not without limitations. First, although this study explored all peer-reviewed articles published in the Scopus database, there is a possibility that there are other related articles that are not indexed in the database and thus were not able to be considered in this study. Therefore, if these articles are available and accessible, future study may consider including other top databases such as Web of Science, JSTOR, and ProQuest as well in the study. Second, although this study followed the PRISMA protocol and PSALSAR framework to minimize subjectivity in selecting and reviewing the articles based on the defined and categorized vulnerable groups and keywords, bias may not have been fully accounted for during the search, selection and content analysis process. Therefore, future research may consider expanding the search with more keywords and using other analysis methods.

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Author contributions
C.T. Lye: Study conception, supervision, analysis, and the original draft of the manuscript. L.Y. Tay: Study conception and design, methodology, review and editing. T.H. Ng: Conceptualization, material preparation, review and editing. A.H. Ahmad Nazmi: Material preparation, data collection and analysis.

Consent to publish
All authors have read and approved the submission of the final manuscript.

Competing interests
The authors declare no competing interests.

Data availability
Not applicable.

Clinical trial number, Consent to participate and **Ethics approval**
Not applicable.

APPENDICES

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Appendix A Summary of theoretical frameworks used in fintech adoption research

Theoretical Framework/Model	Research Focus	Country	Vulnerable Group	Reference
TAM	Fintech services adoption; mobile wallet adoption; digitized financial services adoption; e-wallet adoption; P2P lending intention.	China, Indonesia, India, Oman, Ghana, Malaysia	Rural, Women, Elderly, Children and young people	[142]; [143]; [179]; [181]; [187]; [189]; [190]; [194]; [216]; [235]
TAM, Diffusion of Innovation Theory, Consumer Behavioural Theory	Determinants of fintech adoption in India.	India	Women	[236]
TAM, Social Influence Theory	Effect of various social dynamics and digital literacy on elderly engagement with m-banking in China.	China	Elderly	[204]
TAM, TPB	Determinants of consumers' perceptions and intention to continue using digital financial services.	Vietnam	Children and young people	[188]
TPB	Fintech use; cryptocurrencies adoption; digital financial literacy	Bangladesh, Spain, India.	Women	[122]; [134]; [141]
TPB, Bottom of Pyramid Theory	Examine effect of knowledge of inclusive finance on Chinese farmers' financing intention.	China	Rural	[186]
TPB, Social Role Theory	Effect of digital financial literacy on cashless payment systems among university students.	Jordan	Children and young people	[193]
TPB, Antecedent-Behaviour-Consequence Model, and Gender Schema Theory	Examine continuous use of fintech in Ghana.	Ghana	General vulnerable groups	[63]
UTAUT	Fintech use; digital finance platform usage; fintech services adoption intention	Brazil, China, Indonesia	Rural, Women, Children and young people	[128]; [178]; [191]
UTAUT, Social Cognitive Theory	Determinants of ZakaTech adoption in Saudi Arabia.	Saudi Arabia and Egypt	Women	[139]
UTAUT2	Automated banking services usage; mobile banking for zakat; sustainable digital financial inclusion.	Colombia, Malaysia, India.	Poverty, Women	[77]; [86]; [234]
UTAUT2, Appraisal Tendency Framework.	Effect of mobile money on access and use of financial services in Uganda during pandemic.	Uganda	Poverty	[95]
Cognitive, Affective, and Normative Model	Factors influencing digital insurance adoption by persons with disabilities in India.	India	People with disabilities	[201]
Diffusion of Innovation Theory	Effect of fintech on financial inclusion in rural Tanzania.	Tanzania	Rural	[182]

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Appendix B List of relevant references in fintech, financial inclusion and general vulnerable groups

Reference	Research Objective	Country, Data and Sample	Research and Data Analysis Methods
[46] Zhao et al. (2024)	Effect of mobile payments on happiness of Chinese people.	China; CHFS; 2017.	Quantitative (Secondary Data); Ordered Probit Regression Model
[48] Pradhan et al. (2024)	Effect of fintech on financial inclusion in India.	India; Questionnaire; 104 respondents.	Quantitative (Primary Data); Logistic and Poisson Regression Models
[49] Orazi et al. (2024)	Socioeconomic determinants of fintech in Argentina.	Argentina; Survey of the Project of the Executive Unit; 2021.	Quantitative (Secondary Data); Probit Regression Model
[50] Elouaourti and Ibourk (2024)	Fintech risks and opportunities in the MENA.	MENA region; Global Findex; 2021.	Quantitative (Secondary Data); Probit Model

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[51] Khan and Alhadi (2022)	Determinants of financial inclusion, saving and credit through the formal financial sector in Saudi Arabia.	Saudi Arabia; World Bank's Global Financial Development Database; 2017	Quantitative (Primary Data); Probit Regression Model
[52] Kodongo (2024)	Effect of fintech on the use of traditional financial products.	Kenya; FinAccess Kenya Household Survey; 2021.	Quantitative (Secondary Data); Poisson, Logistic, Selection Model, Lewbel Instrumental Variable (IV), Linear Probability Model, and Recentered Influence Functions Treatment-effects Regression Models
[53] Ahamadou and Agada (2023)	Effect and determinants of fintech on financial inclusion.	The WAEMU countries except Guinea-Bissau; Global Findex; 2017.	Quantitative (Secondary Data); Logit Regression Model
[54] Mogaji and Nguyen (2022)	Examine the dark side of mobile money in SSA.	SSA; Interview; 41 financially vulnerable customers, 19 mobile money agents, and 11 fintech developers.	Qualitative; Thematic Analysis
[57] Koomson et al. (2023)	Effect of mobile money on entrepreneurship.	Kenya, Tanzania and Uganda; InterMedia Financial Inclusion Insights survey; 2017.	Quantitative (Secondary Data); Lewbel 2SLSs Regression Model
[58] Chen et al. (2024)	Effect of fintech and mobile banking adoption on village and township banks.	China; Questionnaire and Tianyancha Enterprise Database; 900 village and township banks.	Quantitative (Primary Data); Regression and IV Models
[59] Jia and Kanagaretnam (2025)	Examine the link between digital inclusion, social and ethical challenge of financial inclusion.	U.S.; Prosper Marketplace P2P data, and Call Reports databases; 2009-2020.	Quantitative (Secondary Data); Difference-in-Differences, Multivariate, and System GMM Models
[60] Okello Candiya Bongomin et al. (2025)	Effect of digital literacy and fintech on financial inclusion of vulnerable groups.	Uganda; Questionnaire; 330 respondents from vulnerable groups.	Quantitative (Primary Data); Covariance-Based SEM
[61] Katyayani and Varalakshmi (2019)	Effects of demographic factors on fintech adoption in India.	India; Questionnaire; 100 respondents.	Quantitative (Primary Data); Analysis of Variance
[62] Ha et al. (2025)	Determinants of digital financial inclusion in ASEAN.	ASEAN countries; Global Findex; 2011, 2014, 2017, and 2021.	Quantitative (Secondary Data); Logit Regression Model
[63] Hanu et al. (2024)	Examine continuous use of fintech in Ghana.	Ghana; Questionnaire, 1292 respondents; Interview 28 interviewees.	Mixed-Method; PLS-SEM
[64] Yang et al. (2023)	Effect of financial literacy on digital finance in China.	China; CHFS; 2015, 2017.	Quantitative (Secondary Data); Probit with IV Regression Model
[65] Kass-Hanna et al. (2022)	Effect of financial and digital literacy on financial resilience.	7 countries; InterMedia Financial Inclusion Insights survey; 2017.	Quantitative (Secondary Data); Probit Regression, 2SLS, and Linear Probability Models
[66] Nandru et al. (2023)	Indian socioeconomic effect on fintech during the COVID-19 pandemic.	India; Global Findex 2021. 3,000 respondents aged 15 and above.	Quantitative (Secondary Data); Logistic Regression Model

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Appendix C List of relevant references in fintech, financial inclusion and poverty

Reference	Research Objective	Country, Data and Sample	Research and Data Analysis Methods
[39] Demir et al. (2022)	The relationships between fintech, financial inclusion and income inequality.	140 countries; The Standardized World Income Inequality, Global Financial Inclusion, and WDI Databases 2011, 2014 and 2017.	Quantitative (Secondary Data); Quantile Regression Model
[70] Fu and Liu (2023)	Effect of financial service accessibility on income inequality.	The World Bank's Development Research Bureau, and Standardized World Income Inequality Databases; 2004-2018.	Quantitative (Secondary Data); Benchmark Regression Model
[71] Chinoda and Mashamba (2021)	Mediating effect of financial inclusion on fintech and income inequality.	25 African countries; 2011, 2014, and 2017.	Quantitative (Secondary Data); SEM
[72] Nasution et al. (2022)	Effect of financial inclusion and fintech on economic and poverty of ASEAN-5 countries.	ASEAN-5 (Indonesia, Philippines, Vietnam, Malaysia, Thailand); The World Bank Database; 2009-2019 .	Quantitative (Secondary Data); Seemingly Unrelated Regression (SUR) Model with Feasible

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			Generalized Least Square (FGLS) Estimation.
[73] Wong et al. (2023)	Effect of financial inclusion on poverty and income inequality.	ASEAN-5 (Indonesia, Philippines, Vietnam, Malaysia, Thailand); IMF Financial Access Survey Database, World Bank Database, the Standardized World Income Inequality Database, and Asian Development Bank's Database; 2008-2019.	Quantitative (Secondary Data); CS-ARDL Model.
[74] Erel and Liebersohn (2022)	Effect of fintech on financial services and inclusion of small businesses.	U.S.; Small Business Administration PPP Loan Database, and American Community Survey 2014–2018.	Quantitative (Secondary Data); Fixed-Effect Regression Model
[75] Ahmad and Wongsuraw at (2023)	Effect of financial innovative products on welfare of the Bangladeshi low-income households.	Bangladesh; Questionnaire; 398 textile factory workers.	Quantitative (Primary Data); Probit Regression Model
[76] Kim (2020)	Effect of mobile money on urban poor financial inclusion.	Kenya; Questionnaire and Kenya FinAccess survey data 2016; 358 respondents.	Quantitative (Primary Data); McNemar Test
[77] Correa et al. (2022)	Disposition of low-income Colombians using automated banking services.	Colombia; Questionnaire; 483 low-income residents.	Quantitative (Primary Data); SEM
[78] Liu and Guo (2023)	Effect of digital financial inclusion on poverty.	China; CHFS database 2015, 2017, and 2019; 29,625 households.	Quantitative (Secondary Data); Fixed Effects, IV, and Difference-in-Differences Models.
[80] Liu et al. (2023)	Effect of fintech on income inequality.	China; China Inter-Provincial panel data, and DFI Index Database; 2011-2020.	Quantitative (Secondary Data); GMM, Panel Threshold and Quantile Regression Models.
[81] Tao et al. (2023)	Effect of digital inclusive finance on poverty.	China; CHFS database 2017, 2019, and 2021; 47,263 households.	Quantitative (Secondary Data); Probit Regression Model
[82] Chen et al. (2022)	Effect of digital technologies on poverty in rural China.	China; CFPS and DFI Index (2011-2018).	Quantitative (Secondary Data); Logistic Regression Model
[83] Wang and Fu (2022)	Effect of digital financial inclusion on poverty in rural China.	China; China Labour Force Dynamics Survey database, and the DFI Index database; 4,326 households.	Quantitative (Secondary Data); Probit Regression Model, and Stochastic Frontier Analysis.
[84] Adbi and Natarajan (2023)	Effect of mobile money and bank account ownership on savings for micro-enterprises.	74 low- and middle-income countries; The Global Findex 2017.	Quantitative (Secondary Data); Logistic Regression Model
[85] Aderemi and Ishak (2023)	Examine if Qard Hasan can be an Islamic crowdfunding in Malaysia.	Malaysia; Semi-structured interview; 6 Interviewees.	Qualitative; Thematic Analysis
[86] Ahmad and Yahaya (2023)	Determinants of mobile banking for zakat distribution among *asnaf* in Malaysia.	Malaysia; Questionnaire; 470 *asnaf*s.	Quantitative (Primary Data); SEM
[87] Mushtaq and Bruneau (2019)	Effect of ICT on financial inclusion and poverty in developing countries.	62 Countries; Microfinance Information Exchange, WDI, and International Telecommunication Union Database 2015; 2001–2012.	Quantitative (Secondary Data); Pooled OLS, Fixed- and Random-Effects, Quintile and IV, and GMM Models.
[89] Collins et al. (2023)	The use and perceptions of financial products among low-income parents.	U.S.; FDIC Survey 2019.	Mixed-Method; Thematic Analysis
[91] Yakubi et al. (2022)	Effect of regulations and fintech on financial inclusion and development.	77 low-income countries; IMF, UNDP and World Bank Databases.	Quantitative (Secondary Data); PLS-SEM
[93] Wang et al. (2022)	Effect of financial inclusion on poverty in rural China.	China; CBIRC, CHFS, the China Statistical Yearbook, and the DFI Index Databases; 39739 households.	Quantitative (Secondary Data); Quantile Regression, IV, and Fixed-Effects Models.
[94] Billore and Billore (2020)	Fintech usage among low-income customers in India during pandemic.	India; Questionnaire; 120 respondents.	Quantitative (Primary Data); Regression Model
[95] Bongomin et al. (2023)	Effect of mobile money on access and use of financial services in Uganda during pandemic.	Uganda; Questionnaire; 2737 respondents.	Quantitative (Primary Data); PLS-SEM

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[96] Emara (2023)	Effect of fintech on poverty in SSA region.	SSA region; 2004-2020.	Quantitative (Secondary Data); GMM Model
[97] Hu et al. (2023)	Effect of digital finance and entrepreneurship on household income disparity.	China; DFI Index and CFPS Databases; 2014, 2016, and 2018.	Quantitative (Secondary Data); IV Regression Model
[101] Danisewicz and Elard (2023)	Effect of fintech on personal bankruptcy.	U.S.; Federal Judicial Centre, and New York Federal Reserve Center Databases.	Quantitative (Secondary Data); Difference-in-Differences Model
[102] Hunter et al. (2024)	Effect of mobile banking on cash demand in Bangladesh and Indonesia.	Indonesia and Bangladesh; Questionnaire; 2000 respondents from low-income households.	Quantitative (Primary Data); OLS and Probit Regression Models
[106] Duan et al. (2024)	Effect of fintech development on urban poverty.	China; CFPS and DFI Index; 2012-2020.	Quantitative (Secondary Data); OLS and Logit Regression Models
[107] Suhrab et al. (2024)	Effect of digital financial inclusions on income inequality.	BRICS; World Income Inequality, Global Findex, WDI, and OECD Statistics Databases; 2000-2019.	Quantitative (Secondary Data); Driscoll-Kraay and Fixed-effects Regression Model
[239] Fontin and Lin (2019)	The levels and determinants of financial innovation.	9 low-income countries; 168 Banks.	Quantitative (Secondary Data); DEA and SFA

Appendix D List of relevant references in fintech, financial inclusion and women

Reference	Research Objective	Country, Data and Sample	Research and Data Analysis Methods
[110] Esmaeilpour Moghadam and Karami (2023)	Effect of fintech and financial inclusion on women's empowerment.	113 countries; the World Bank, and Global Findex Databases 2017.	Quantitative (Secondary Data); Regression Model
[111] Tripathi and Rajeev (2023)	Develop and evaluate gender-based financial inclusion performance of countries worldwide.	109 Countries; Global Findex; 2011, 2014, 2017, and 2021.	Quantitative (Secondary Data); Feasible Generalized Least Squares, and 2SLSs Models
[112] Razzaq et al. (2024)	Determinants of gender gaps in financial inclusion.	Pakistan; Global Findex; 2021.	Quantitative (Secondary Data); Logistic Regression Model
[113] Adeleke and Alabede (2022)	Factors and geographic clustering of financial inclusion.	Nigeria; Enhancing Financial Innovation and Access Database, and National Bureau of Statistics Database; 27,470 Nigerian adults (18 years above).	Quantitative (Secondary Data); Regression Models
[114] Al Khub et al. (2024)	Effect of digital financial inclusion on poverty and economic growth.	Jordan; Questionnaire; 260 respondents.	Quantitative (Primary Data); Non-linear Probit Regression Model
[115] Guo et al. (2021)	Effect of fintech on gender wage gap.	China; DFI Index, Chinese General Social Survey, China Statistical Yearbook and China Regional Financial Operation Report databases; 2012, 2013, and 2015.	Quantitative (Secondary Data); OLS and Probit Regression Models
[116] Han et al. (2023)	Effect of digital finance on women's bargaining power.	China; CFPS, and DFI Index databases; 201-2018.	Quantitative (Secondary Data); OLS and Logit Regression Models
[117] Chen et al. (2023)	Examine gender differences in fintech adoption.	28 Countries; EY Global Fintech Adoption Index Database 2019.	Quantitative (Secondary Data); Fixed-Effect Regression
[118] Chatterjee (2024)	Investigate the regional disparity in digital and gender divide within developing countries.	123 economies; Gallup World Poll data; 2021.	Quantitative (Secondary Data); IV and Bivariate Probit Models
[119] Yeyouomo et al. (2023)	Effect of fintech on financial inclusion gender gap in SSA.	27 SSA Countries; Global Findex, WDI Database, and Heritage Foundation Database; 2011, 2014 and 2017.	Quantitative (Secondary Data); Tobit Regression Model
[120] Makhkamova and Saidmurodov (2019)	Investigate fintech usage and financial inclusion of women in financial market.	Tajikistan; Questionnaire; 105 respondents.	Descriptive; Descriptive Analysis
[121] Noreen et al. (2022)	Effect of socio-demographic characteristics on mobile money usage in Pakistan.	Pakistan; Questionnaire; 436 respondents.	Descriptive; Descriptive Analysis

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[122] Hoque et al. (2024)	Effect of sociodemographic, social and facilitating influences, on fintech use.	Bangladesh; Questionnaire; 237 participants.	Quantitative (Primary Data); Ordered Logit Regression and Blinder-Oaxaca Decomposition Models
[123] Özşuca (2024)	Determinants of digital finance access between genders in developing countries.	81 low- and middle-income countries; Global Findex; 2021.	Quantitative (Secondary Data); Probit and Fairlie Decompositions Models
[124] Song et al. (2024)	Effect of fintech on bank loan gender gap.	U.S.; 60 U.S. banks; 2012-2022.	Quantitative (Secondary Data); Two-step System GMM and Difference in differences approaches.
[127] Kedir and Kouame (2022)	Effect of fintech on poverty and inequality among women in Burkina Faso and Cameroon.	Cameroon and Burkina Faso; FinScope Consumer Surveys Database; 2016, 2017.	Quantitative (Secondary Data); Probit Regression Model
[128] Kurniasari and Lestari (2024)	Effect of financial literacy on women fintech use and business success.	Indonesia; Questionnaire; 270 women entrepreneurs.	Quantitative (Primary Data); PLS-SEM
[129] Dura and Wardana (2024)	Effect of fintech and financial literacy on women entrepreneurs in Indonesia.	Indonesia; Questionnaire; 390 female entrepreneurs.	Quantitative (Primary Data); Multiple Linear Regression Model
[130] Ndione et al. (2024)	Effect of banks, microfinance institutions, and e-money on financial inclusion.	8 WAEMU countries; The West African States Central Bank and the WGI databases; 2010 to 2020.	Quantitative (Secondary Data); Regression Model
[131] Fisch et al. (2022)	Effect of gender, ethnicity and geography on the ICO choice.	Cross-country; Crunchbase, the World Bank, and OECD databases; ICO ventures, VC-backed blockchain ventures, and real estate investment trusts (REITs) databases.	Quantitative (Secondary Data); Probit, Tobit, and Instrumental Regression Models
[132] Cicchiello and Kazemikhasragh (2022)	Examine gender-related differences among investors in equity seeking firms.	Latin America (Brazil, Chile and Mexico); 492 projects between 2013 and 2017.	Quantitative (Secondary Data); Poisson Regression
[133] Kim (2022)	Effect of mobile money on financial inclusion of women.	Kenya; Mixed-Method; Questionnaire (341 respondents excluding teenagers), Interview (72 participants).	Mixed-Method; McNemar's test
[135] Chutipat et al. (2023)	Determinants of adoption of utility tokens in Thailand.	Thailand; Questionnaire; 1,280 Thai investors.	Quantitative (Primary Data); Binary Logistic Regression Model
[136] Raheem et al. (2024)	Effect of fintech on financial inclusion of women.	Ghana; Interview and Observation; 154 respondents.	Qualitative; Thematic and Iterative data analysis
[137] Imam et al. (2022)	Opportunities and challenges of fintech in the ASEAN and SAARC regions.	ASEAN and SAARC countries; the Global Findex 2017.	Quantitative (Primary Data); OLS and Fixed-Effect Regression Models
[138] Baber (2020)	Comparison of fintech and financial inclusion performances of countries adopting Islamic and conventional finance systems.	20 Countries; the Global Findex 2017, and Islamic Finance Country Index Database 2017.	Quantitative (Secondary Data); t-test
[139] Bin-Nashwan et al. (2023)	Determinants of ZakaTech adoption in Saudi Arabia.	Saudi Arabia and Egypt; Questionnaire; 1,006 zakat payers.	Quantitative (Primary Data); PLS-SEM
[140] Hasan et al. (2023)	Effect of digital financial literacy on female entrepreneurs' financial inclusion.	144 countries; the Global Findex 2017.	Quantitative (Secondary Data); Probit and Tobit Regression Models
[141] Mishra et al. (2024)	Effect of digital financial literacy on financial decision-making among women in India.	India; Questionnaire; 385 women.	Quantitative (Primary Data); SEM
[142] Setiawan et al. (2023)	Determinants of fintech adoption intention of women in Indonesia.	Indonesia; Questionnaire; 409 female respondents.	Quantitative (Primary Data); PLS-SEM
[143] Igamo et al. (2024)	Determinants of fintech adoption intention and usage.	Indonesia; Questionnaire; 403 women respondents.	Quantitative (Primary Data); PLS-SEM
[144] Nguyen (2022)	Determinants and effect of financial knowledge on fintech usage in Vietnam.	Vietnam; Questionnaire; 527 respondents.	Quantitative (Primary Data); Instrumental Regression and 2SLS Models

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[145] Belgacem et al. (2024)	Effect of fintech innovation, demographic, psychological and socio-economic factors on Saudi women's financial literacy.	Saudi Arabia; Questionnaire; 498 female employees.	Quantitative (Primary Data); Regression Model
[146] Omotubora (2024)	Effect of AI on women's financial exclusion in Nigeria.	Nigeria; Survey; 169 licensed fintech providers.	Quantitative (Primary Data); Descriptive
[196] Gonzalez (2023)	Effect of gender, financial literacy, collateral perception, and identification with borrowers on pro-social lending decisions.	U.S.; 663 lending decisions; 221 finance students; Jan 2019 - Mac 2020.	Experimental; Multivariate Analysis
[234] Gupta and Kiran (2024)	Examine the sustainable digital financial inclusion of women in banking and IT sectors.	India; Questionnaire; 257 female bank employees.	Quantitative (Primary Data); PLS-SEM
[235] Kumar and Rani (2025)	Determinants of fintech services adoption in India.	India; Questionnaire; 411 male and 473 female participants.	Quantitative (Primary Data); PLS-SEM
[236] Rani and Kumar (2023)	Determinants of fintech adoption in India.	India; Questionnaire; 333 males and 317 females.	Quantitative (Primary Data); PLS-SEM
[237] Sanad and Al Lawati (2023)	Effect of fintech on gender diversity and firm performance.	Bahrain; Bahrain Bourse website and annual reports; 2016-2022.	Quantitative (Secondary Data); Fixed-Effect Regression Model
[238] Wahbi et al. (2023)	Examine the preferences for digital and conventional credit between genders.	Madagascar; Questionnaire; 420 smallholder farmers.	Quantitative (Primary Data); Logit Regression Model

Appendix E List of relevant references in fintech, financial inclusion and rural communities

Reference	Research Objective	Country, Data and Sample	Research and Data Analysis Methods
[147] Deng et al. (2023)	Effect of digital economy development on urban-rural income gap.	China; DFI Index database; 2011-2019.	Quantitative (Secondary Data); Regression and IV Regression Models
[149] Friedline et al. (2020)	Investigate the relationships between poverty rates, races and fintech rates.	U.S.; FDIC 2014, NCUA 2014, and ACS 2010-2014.	Quantitative (Secondary Data); Regression Models
[152] Wang et al. (2024)	Effect of fintech on farmers' prosperity in China.	China; China Urban Statistical Yearbook; 2013-2018.	Quantitative (Secondary Data); Regression Model
[153] Xia and Kong (2024)	Effect of digital finance on rural revitalization.	China; National Bureau of Statistics of China; 2012-2019.	Quantitative (Secondary Data); Fixed-effects Regression and Differential GMM Models.
[154] Cai et al. (2024)	Effect of fintech adoption on economic development in rural China.	China; China's National Fixed-Point Survey, and DFI Index; 2014-2017.	Quantitative (Secondary Data); Fixed-Effect and IV Regression Models
[155] Song et al. (2020)	Effect of finance and digital finance accessibility on consumption.	China; CHFS, and China Banking Regulatory Commission databases; 2015.	Quantitative (Secondary Data); Regression Models
[156] Zhang et al. (2022)	Effect of mobile payment on rural household consumption.	China; CHFS; 2017.	Quantitative (Secondary Data); 2SLSs Regression Model
[157] Hsueh et al. (2024)	Effect of digital financial development on decision of Chinese rural-urban migrants to settle down.	China; CMDS; 2014–2018.	Quantitative (Secondary Data); Probit Regression Model
[158] Fang (2023)	Effect of digital finance on individual charitable giving in China.	China; CFPS, and DFI Index databases; 2018	Quantitative (Secondary Data); Regression Models
[159] Tetteh (2023)	Effects of digital lending on deprivation in rural Kenya.	Kenya; FinAccess surveys; 2016 and 2019.	Quantitative (Secondary Data); Probit Regression Model
[160] Atta-Ankomah et al. (2024)	Effect of digital finance on rural households' livelihood diversification in Ghana.	Ghana; Ghana Living Standards Surveys 7; 2016-2017.	Quantitative (Secondary Data); OLS, Logit, Poison, and IV Models
[161] Su et al. (2021)	Effect of online transactions on farmers' digital financial inclusion.	China; Questionnaire; 832 respondents.	Quantitative (Primary Data); IV Regression Model
[162] Hao et al. (2023)	Effect of digital finance on urban-rural integration.	China; DFI Index database; 2011-2020.	Quantitative (Secondary Data); GMM

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[163] Ji et al. (2021)	Effect of digital inclusive finance on urban-rural income gap.	China; DFI Index and China City Statistical Yearbook databses; 2014-2018.	Quantitative (Secondary Data); Fixed-Effect Regression
[164] Wang (2024)	Effect of digital financial capability and its influencing mechanisms on entrepreneurship in China.	China; CHFS; 2015, 2017 and 2019.	Quantitative (Secondary Data); Probit Regression with Two-way Fixed-effects Model
[165] Yu et al. (2024)	Effect of digital inclusive finance on entrepreneurship in in rural China.	China; CHFS; 2015, 2017, and 2019.	Quantitative (Secondary Data); 2SLSs with IV
[166] Chai et al. (2023)	Effect of digital finance on entrepreneurs' rate of return in China.	China; DFI Index and CHFS databases; 2015, 2017, and 2019.	Quantitative (Secondary Data); Fixed-Effect Regression
[167] Xu et al. (2024)	Effect of digital finance on rural revitalization in China.	China; DFI Index; 2011-2020.	Quantitative (Secondary Data); Threshold Regression, Heterogeneity Regression, and Dynamic System GMM Models.
[168] Yang et al. (2024)	Effect of digital finance on intergenerational income mobility in China.	China; DFI Index and CFPS; 2014, 2016, and 2018.	Quantitative (Secondary Data); OLS, IV and Tobit Regression Models.
[171] Bi (2024)	Effects of digital finance and infrastructure on development in rural China.	China; National Bureau of Statistics of China, DFI Index, and China Research Data Services Platform; 2003-2021.	Quantitative (Secondary Data); Regression Models
[172] Niu et al. (2022)	Effect of broadband infrastructure on digital financial inclusion.	China; China's Ministry of Industry and Information Technology, National Bureau of Statistics of China, China Population Census 2010, and DFI Index databases; 2014 to 2018.	Quantitative (Secondary Data); Difference-in-Differences Regression Model
[173] Jamil et al. (2024)	Effect of digital inclusive finance on financial inclusion in China.	China; DFI Index, China Rural, China, and Chinese Employment Statistics Yearbooks; 2011-2020.	Quantitative (Secondary Data); Spatial Autocorrelation Analysis, OLS, 2SLS, LIML, GMM, and IGMM Models.
[174] Song et al. (2023)	Effect of mobile payment on the access to traditional bank services in China.	China; Questionnaire; 128 villages.	Quantitative (Primary Data); Hurdle Models with Probit, OLS, and 2SLS.
[175] Munyegera and Matsumoto (2018)	Effect of financial innovation on financial behaviours of rural households in Uganda.	Uganda; Mobile Money follow up survey 2014; 816 rural households.	Quantitative (Primary Data); Probit and Tobit Regression Models
[176] Maskara et al. (2021)	Effect of P2P platforms on financial inclusion.	U.S.; Prosper database; 15 Apr 2008 to 31 Oct 2019.	Quantitative (Secondary Data); Logit Regression Model
[178] Wu et al. (2022)	Factors and effects of digital finance platform usage on personal finance management of rural people.	China; Questionnaire; 403 respondents.	Quantitative (Primary Data); PLS-SEM
[179] Wu and Peng (2024)	Factors of fintech adoption among rural peoples.	China; Questionnaire; 386 rural residents.	Quantitative (Primary Data); SEM
[180] He and Li (2020)	Effect of social interaction on fintech participation among rural households.	China; Rural Inclusive Finance Survey 2017; 1,953 respondents.	Quantitative (Secondary Data); Probit and Count Regression Models
[181] Ghouse et al. (2024)	Determinants of mobile wallet adoption among rural millennials.	Oman; Questionnaire; 544 respondents.	Quantitative (Primary Data); PLS-SEM
[182] Koloseni and Mandari (2024)	Effect of fintech on financial inclusion in rural Tanzania.	Tanzania; Questionnaire; 275 respondents.	Quantitative (Primary Data); SEM
[183] Hasan et al. (2021)	Effects of financial knowledge on financial and fintech access in rural Bangladesh.	Bangladesh; Questionnaire; 852 respondents.	Quantitative (Primary Data); Logistic, Probit, and Complementary Log-log Regression Models
[184] Hasan et al. (2023)	Effect of financial knowledge on fintech access in rural Bangladesh.	Bangladesh; Questionnaire; 817 respondents.	Quantitative (Primary Data); Logistic, Probit, and Complementary Log–log Regression Models
[185] Xu et al. (2024)	Effect of digital financial literacy on income inequality.	China; CHFS; 2017 and 2019.	Quantitative (Secondary Data); Fixed-effects Regression Model.

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[186] Aisaiti et al. (2019)	Examine effect of knowledge of inclusive finance on Chinese farmers' financing intention.	China; Questionnaire; 160 rural farmers.	Quantitative (Primary Data); Regression Models

Appendix F List of relevant references in fintech, financial inclusion and other minority vulnerable groups

Reference	Vulnerable Group	Research Objective	Country, Data and Sample	Research and Data Analysis Methods
[67] Tai et al. (2024)	Elderly	Effect of digital finance on life satisfaction of the elderly.	China; CHPTS, and Digital Inclusive Financial Development Index databases; 2021.	Quantitative (Secondary Data); Ordered Logit, Ordered Probit, and OLS Models
[68] Ha and Nguyen (2024)	Children and Young People	Effect of fintech and financial literacy on financial inclusion.	Vietnam; Questionnaire; 1288 students.	Quantitative (Primary Data); Logistic; Two-stage IV Regression Models
[187] Ayimah et al. (2024)	Young People	Effect of attitude on digitized financial services adoption and its determinants among semi-urban university youths in Ghana.	Ghana; Questionnaire; 375 university students.	Quantitative (Primary Data); Technology Acceptance Model
[188] Dinh and Thanh (2024)	Young People	Determinants of consumers' perceptions and intention to continue using digital financial services.	Vietnam; Questionnaire; 689 Gen Z consumers.	Quantitative (Primary Data); PLS-SEM
[189] Tiwari et al. (2024)	Children and Young People	Determinants of fintech adoption among college students in India.	India; Questionnaire; 397 college students.	Quantitative (Primary Data); SEM
[190] Yang and Shi (2024)	Children and Young People	Determinants of e-wallet adoption intention among youth.	Malaysia; Questionnaire; 140 respondents.	Quantitative (Primary Data); PLS-SEM
[191] Frare et al. (2023)	Young People	Determinants of fintech services adoption intention among business students in Brazil.	Brazil; Questionnaire; 166 university students.	Quantitative (Primary Data); Mixed-Method
[192] Buenestado-Fernández et al. (2023)	Young People	Investigate the perception, obstacles and training preferences of young people about digital financial literacy and competency on digital financial services.	Mexico; 22 focus groups or 200 young participants.	Qualitative; Deductive Analysis
[193] Shehadeh et al. (2025)	Children and Young People	Effect of digital financial literacy on cashless payment systems among university students.	Jordan; Questionnaire; 418 university students.	Quantitative (Primary Data); PLS-SEM
[194] Ichwan and Kasri (2019)	Young People	Determinants of P2P lending intention among youths in Indonesia.	Indonesia; Questionnaire; 400 youths/millennials.	Quantitative (Primary Data); Logistic Regression and SEM
[195] Gonzalez (2020)	Young People	Impact of blockchain-based trust-enhancing heuristics in monitoring and bad loan recovery.	U.S.; 909 lending decisions on P2P site.	Quantitative (Primary Data); t-test
[197] Kameswaran et al. (2023)	People with Disabilities	Investigate the technology access challenges of people with visual impairments to digital banking.	India; Qualitative; 30 interviewees (20 males and 10 females).	Qualitative; Pooled OLS, Fixed- and Random-Effects, Quintile and IV, and GMM Models
[199] Peprah et al. (2023)	People with Disabilities	Impact of disability on financial service access.	Ghana; Ghana Living Standards Survey; 2016-2017.	Quantitative (Secondary Data); IV Regression Model
[200] Gafoor and Amilan (2024)	People with Disabilities	Effect of fintech on financial wellbeing of persons with disabilities in India.	India; Questionnaire; 205 respondents with disabilities.	Quantitative (Primary Data); PLS-SEM
[201] Gupta et al. (2024)	People with Disabilities	Factors influencing digital insurance adoption by persons with disabilities in India.	India; Questionnaire; 323 respondents with disabilities.	Quantitative (Primary Data); Two-stage Hybrid SEM-Artificial Neural Network
[202] Fan et al. (2024)	Elderly	Effect of digital financial inclusion on consumption expenditure of the elderly.	China; DFI Index and CFPS; 2016, 2018, and 2020.	Quantitative (Secondary Data); OLS and IV Models
[203] Choi et al. (2024)	Elderly	Investigate the relationships between demographic, personal, socioenvironmental, and digital usage characteristics.	South Korean; Korea Media Panel Survey; 2019-2020.	Quantitative (Secondary Data); Logistic and Multiple Linear Regression Models

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[204] Mei (2024)	Elderly	Effect of various social dynamics and digital literacy on elderly engagement with m-banking in China.	China; Questionnaire; 425 elderly participants.	Quantitative (Primary Data); PLS-SEM
[205] Lei et al. (2023)	Elderly	Effect of digital financial inclusion on subjective well-being of older population in China.	China; CHRLS database; 2011, 2013, 2015, 2018.	Quantitative (Secondary Data); Two-way Fixed Effect Model, and IV Estimation
[212] Wang et al. (2024)	Minorities	Effect of digital financial inclusion on herdsmen's income in China.	China; DFI Index, China, and China Rural Statistical Yearbooks; 2011-2022.	Quantitative (Secondary Data); OLS, Random- and Fixed-effects Regression Models
[213] Li et al. (2022)	Migrants	Effect of digital finance on migrant entrepreneurs.	China; CMDS, and DFI Index databases; 2012-2017.	Quantitative (Secondary Data); LPM Model, Logit Model, Multinomial Logit Model
[214] Liu et al. (2022)	Migrants	Effect of digital finance on migrant entrepreneurs.	China; CMDS; 2018.	Quantitative (Secondary Data); Probit model
[215] Zhuo and Jia (2024)	Migrants	Effect of green digital finance development on resident migration decisions.	China; CFPS, and DFI Index; 2012-2020 biannually.	Quantitative (Secondary Data); Logit Regression Model
[216] Anwar et al. (2023)	Elderly	Investigate usage of mobile wallet among digital immigrants in Indonesia.	Indonesia; Questionnaire; 400 mobile wallet users.	Quantitative (Primary Data); PLS-SEM

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